Linear Algebra Workbook Solutions

Transformations

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FUNCTIONS AND TRANSFORMATIONS

■ 1. The transformation T maps every vector in $\mathbb{R}^4$ to $\vec{O} = (0,0,0)$. What are the domain, codomain, and range of T?

Solution:

Because T is mapping vectors in $\mathbb{R}^4$ to vector $\vec{O} = (0,0,0)$ in $\mathbb{R}^3$, we can express T as $T : \mathbb{R}^4 \to \mathbb{R}^3$, and say that the domain of the transformation is $\mathbb{R}^4$ and that the codomain is $\mathbb{R}^3$.

Since the transformation is mapping every vector in $\mathbb{R}^4$ to only the zero vector $\vec{O} = (0,0,0)$ in $\mathbb{R}^3$, the range of T is just the zero vector, $\vec{O} = (0,0,0)$.

■ 2. The transformation T maps every vector in $\mathbb{R}^3$ to every vector in $\mathbb{R}^2$. What are the domain, codomain, and range of T?

Solution:

The domain is the space T is mapping *from*, so the domain is $\mathbb{R}^3$. The codomain is the space T is mapping *to*, so the codomain is $\mathbb{R}^2$. And the range is the specific set of vectors within the codomain that are being mapped to. Because every vector in $\mathbb{R}^2$ is being mapped to, the range is all of $\mathbb{R}^2$.

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■ 3. The transformation T maps every vector in $\mathbb{R}^3$ to the zero vector $\vec{O}$ in $\mathbb{R}^3$. What are the domain, codomain, and range of T?

Solution:

The domain is the space T is mapping *from*, so the domain is $\mathbb{R}^3$. The codomain is the space T is mapping *to*, so the codomain is also $\mathbb{R}^3$. And the range is the specific set of vectors within the codomain that are being mapped to, so the range is $\vec{O} = (0,0,0)$.

■ 4. The transformation T maps $\vec{a} = (-1,0,3)$ to $\vec{b} = (-2,1,-2)$. What are the domain, codomain, and range of T?

Solution:

The domain is the space T is mapping *from*, so the domain is $\mathbb{R}^3$. The codomain is the space T is mapping *to*, so the codomain is $\mathbb{R}^3$. And the range is the specific set of vectors within the codomain that are being mapped to, so the range is the single vector being mapped to, $\vec{b} = (-2,1,-2)$.

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■ 5. The transformation T maps $\vec{a} = (-2,0)$ to every vector in $\mathbb{R}^4$. What are the domain, codomain, and range of T?

Solution:

The domain is the space T is mapping *from*, so the domain is $\mathbb{R}^2$. The codomain is the space T is mapping *to*, so the codomain is $\mathbb{R}^4$. And the range is the specific set of vectors within the codomain that are being mapped to. Because every vector in $\mathbb{R}^4$ is being mapped to, the range is all of $\mathbb{R}^4$.

■ 6. The transformation T maps $\vec{a} = (-2, -3,1)$ to the zero vector $\vec{O}$ in $\mathbb{R}^3$. What are the domain, codomain, and range of T?

Solution:

The domain is the space T is mapping *from*, so the domain is $\mathbb{R}^3$. The codomain is the space T is mapping *to*, so the codomain is also $\mathbb{R}^3$. And the range is the specific set of vectors within the codomain that are being mapped to, so the range is the single vector being mapped to, $\vec{O} = (0,0,0)$.

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TRANSFORMATION MATRICES AND THE IMAGE OF THE SUBSET

■ 1. Find the resulting vector $\vec{b}$ after $\vec{a} = (1,6)$ undergoes a transformation by matrix M.

$$M = \begin{bmatrix} -7 & 1 \\ 0 & -2 \end{bmatrix}$$

Solution:

To apply a transformation matrix to vector $\vec{a}$, we'll multiply the matrix by the vector.

$$\vec{b} = M\vec{a} = \begin{bmatrix} -7 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

$$\vec{b} = M\vec{a} = \begin{bmatrix} -7(1) + 1(6) \\ 0(1) - 2(6) \end{bmatrix}$$

$$\vec{b} = M\vec{a} = \begin{bmatrix} -7 + 6 \\ 0 - 12 \end{bmatrix}$$

$$\vec{b} = M\vec{a} = \begin{bmatrix} -1 \\ -12 \end{bmatrix}$$

■ 2. Sketch triangle $\triangle ABC$ with vertices $(2,3)$, $(-3,-1)$, and $(1,-4)$, and the transformation of $\triangle ABC$ after it's transformed by matrix L.

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$$L = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

Solution:

Put the vertices of $\triangle ABC$ into a matrix.

$$\begin{bmatrix} 2 & -3 & 1 \\ 3 & -1 & -4 \end{bmatrix}$$

Apply the transformation of L to the vertex matrix.

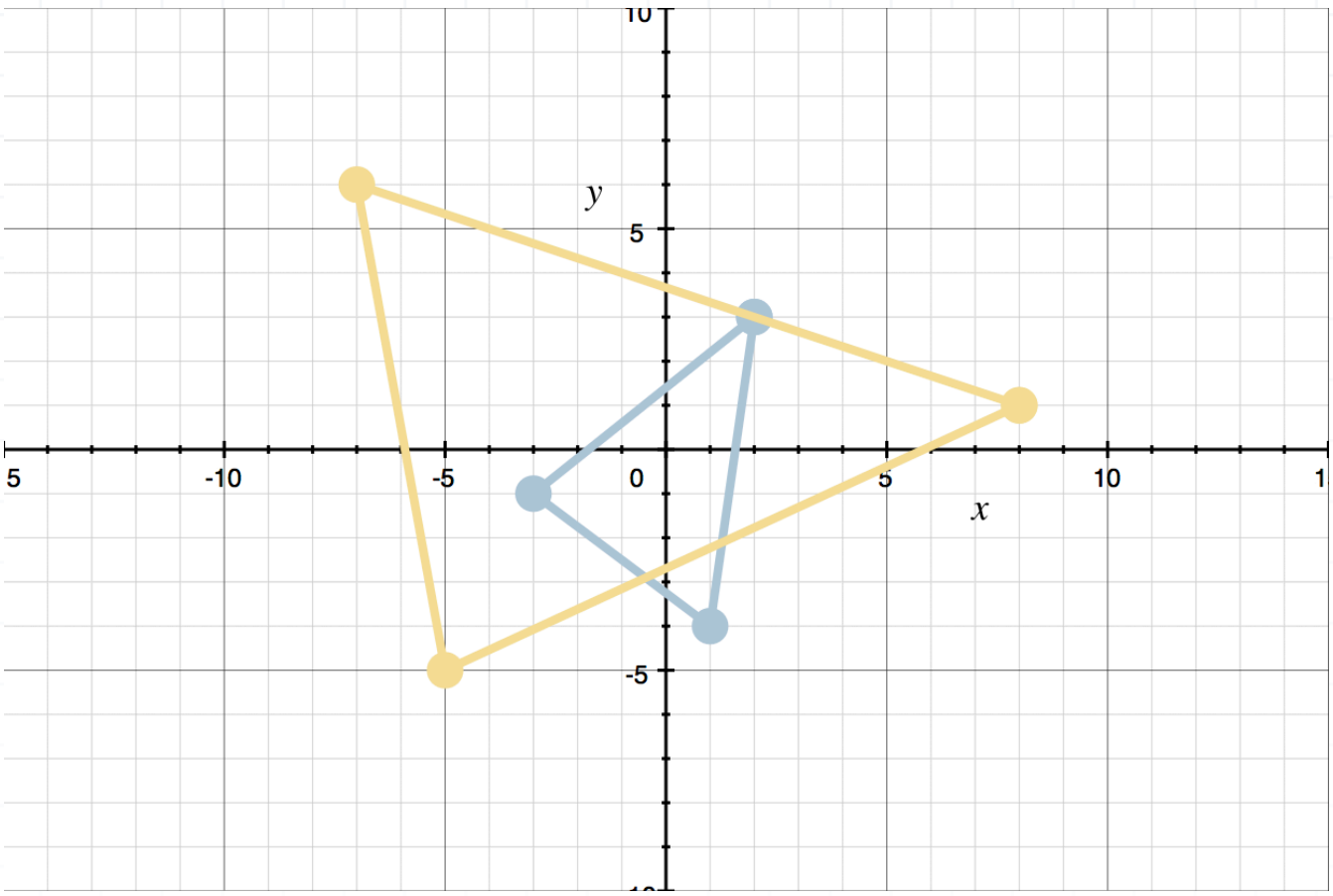
$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -3 & 1 \\ 3 & -1 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1(2)+2(3) & 1(-3)+2(-1) & 1(1)+2(-4) \\ 2(2)-1(3) & 2(-3)-1(-1) & 2(1)-1(-4) \end{bmatrix}$$

$$\begin{bmatrix} 8 & -5 & -7 \\ 1 & -5 & 6 \end{bmatrix}$$

The original triangle $\triangle ABC$ is sketched in light blue, and its transformation after L is in yellow.

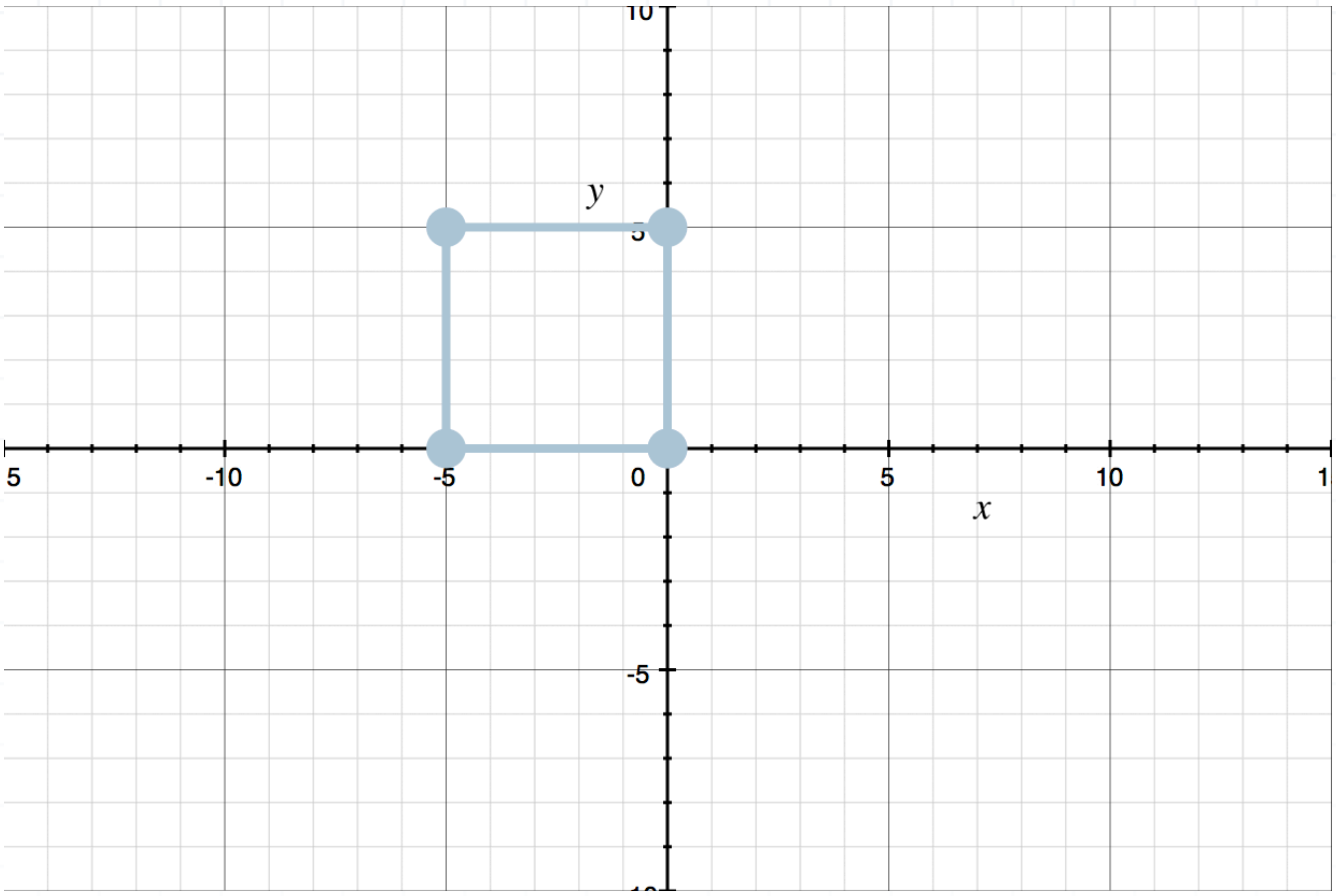
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3. Sketch the transformation of the square in the graph after it's transformed by matrix Z.

$$Z = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$$

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Solution:

Put the vertices of the square into a matrix.

$$\begin{bmatrix} 0 & -5 & -5 & 0 \\ 0 & 0 & 5 & 5 \end{bmatrix}$$

Apply the transformation of Z to the vertex matrix.

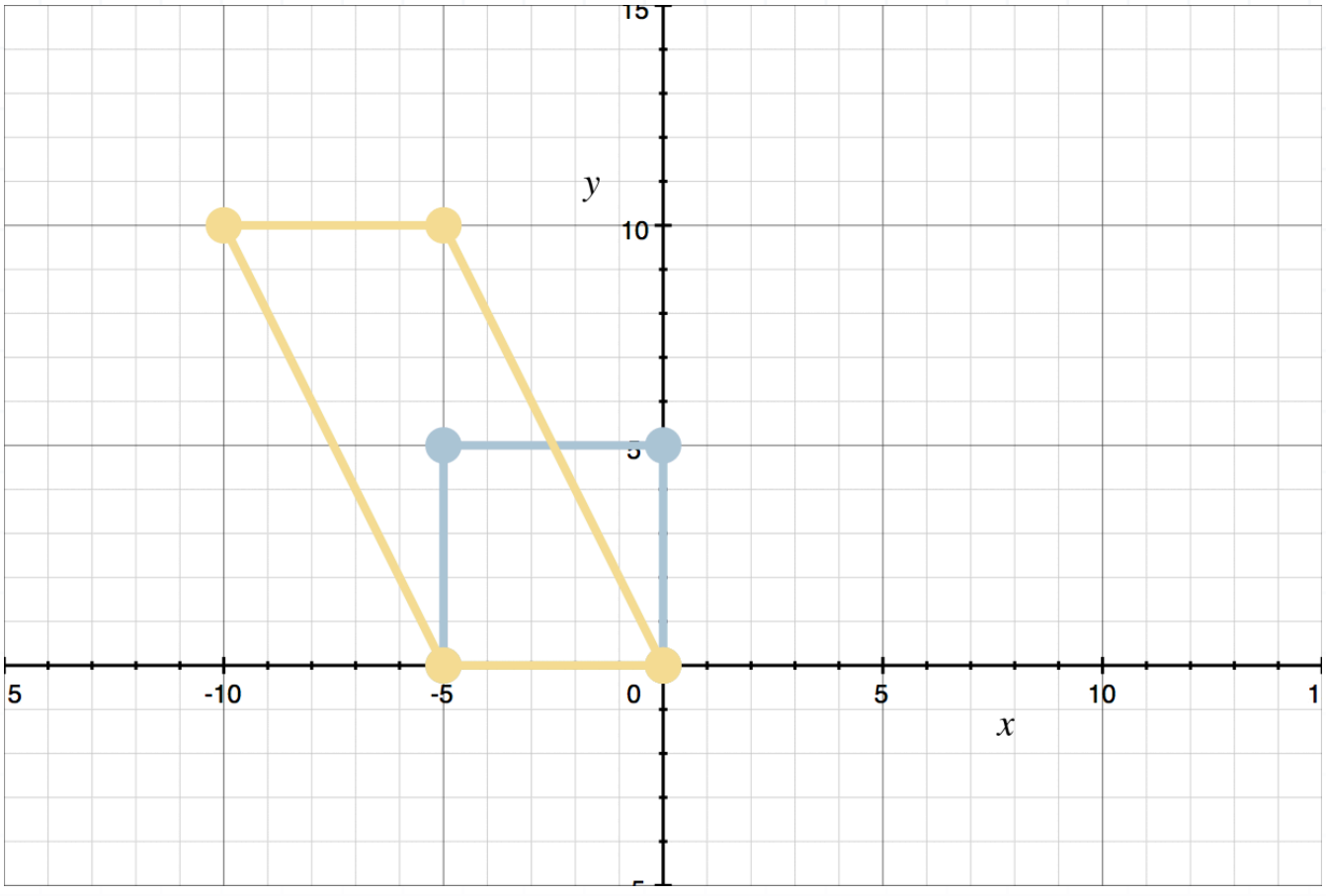
$$\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 & -5 & -5 & 0 \\ 0 & 0 & 5 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1(0)-1(0) & 1(-5)-1(0) & 1(-5)-1(5) & 1(0)-1(5) \\ 0(0)+2(0) & 0(-5)+2(0) & 0(-5)+2(5) & 0(0)+2(5) \end{bmatrix}$$

$$\begin{bmatrix} 0 & -5 & -10 & -5 \\ 0 & 0 & 10 & 10 \end{bmatrix}$$

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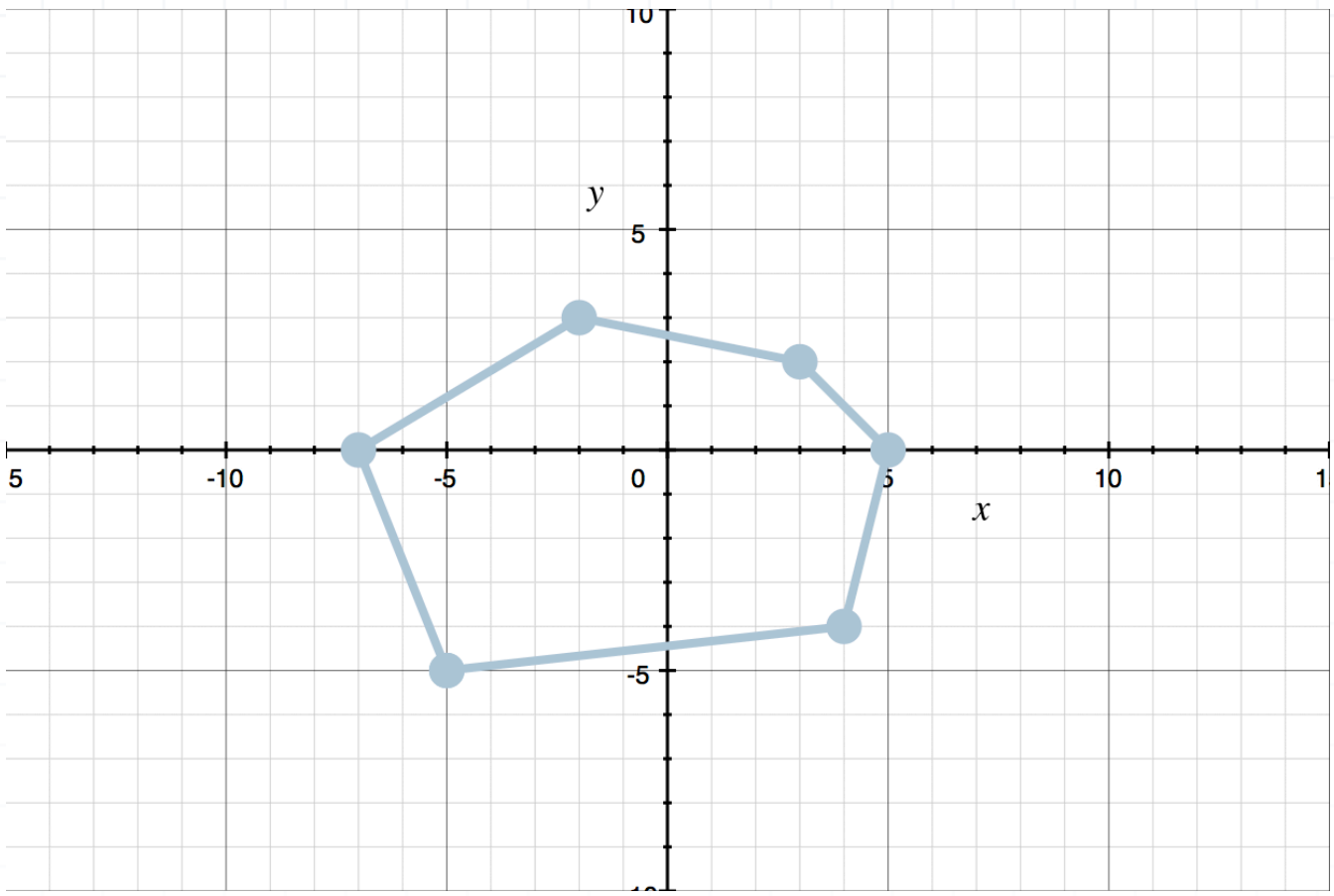
The original square is sketched in light blue, and its transformation after Z is in yellow.



4. Sketch the transformation of the hexagon after it's transformed by matrix Y.

$$Y = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}$$

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Solution:

Put the vertices of the hexagon into a matrix.

$$\begin{bmatrix} -5 & 4 & 5 & 3 & -2 & -7 \\ -5 & -4 & 0 & 2 & 3 & 0 \end{bmatrix}$$

Apply the transformation of Y to the vertex matrix.

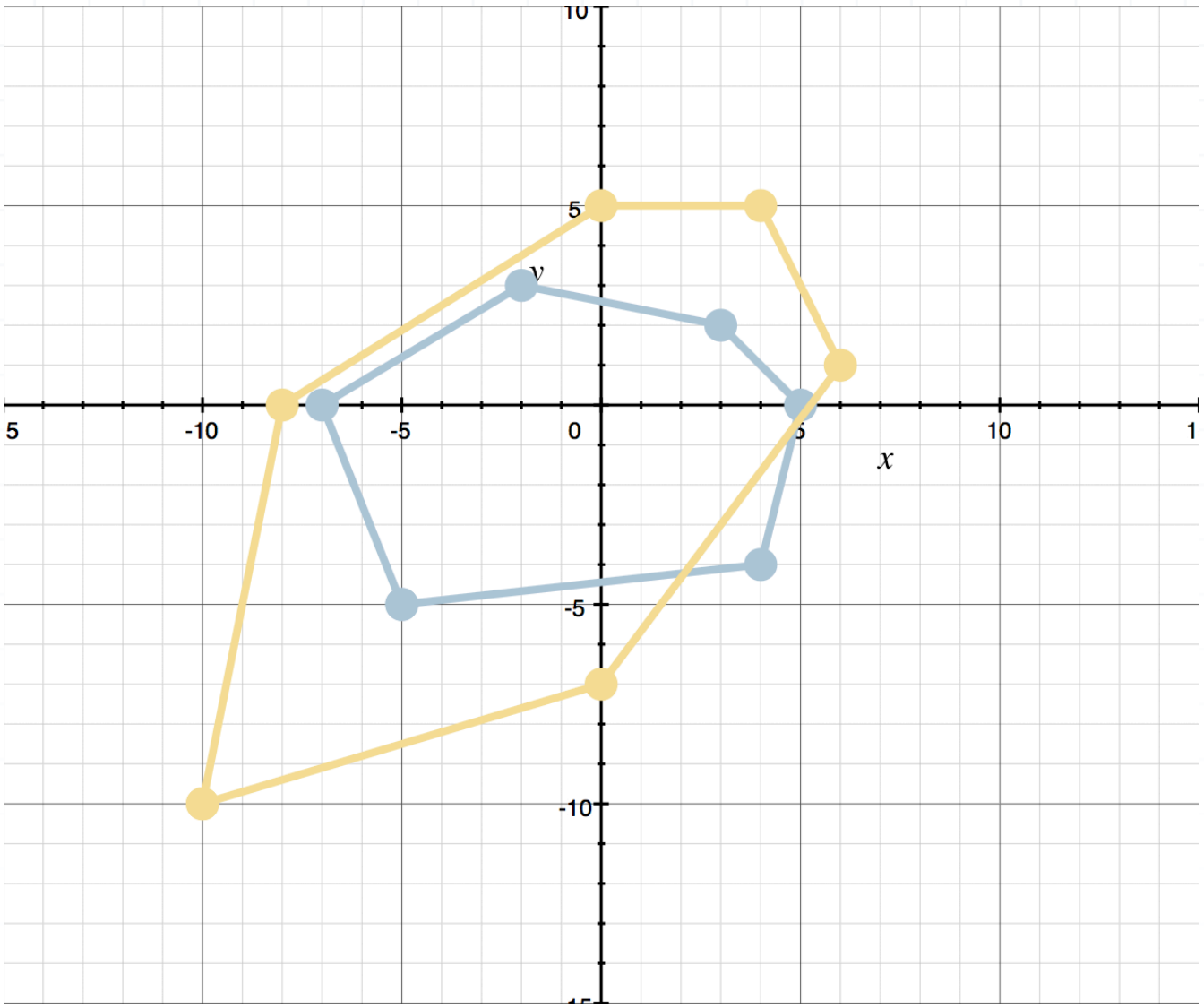
$$\begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -5 & 4 & 5 & 3 & -2 & -7 \\ -5 & -4 & 0 & 2 & 3 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -5(0)-5(2) & 4(0)-4(2) & 5(0)+0(2) & 3(0)+2(2) & -2(0)+3(2) & -7(0)+0(2) \\ -5(1)-5(1) & 4(1)-4(1) & 5(1)+0(1) & 3(1)+2(1) & -2(1)+3(1) & -7(1)+0(1) \end{bmatrix}$$

$$\begin{bmatrix} -10 & -8 & 0 & 4 & 6 & 0 \\ -10 & 0 & 5 & 5 & 1 & -7 \end{bmatrix}$$

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The original hexagon is sketched in light blue, and its transformation after Y is in yellow.



■ 5. What happens to the unit vector $\vec{a} = (1,0)$ after the transformation given by matrix K.

$$K = \begin{bmatrix} 3 & -5 \\ -1 & 0 \end{bmatrix}$$

Solution:

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Where $\vec{a} = (1,0)$ lands is given by the first column of the transformation matrix. So $\vec{a}$ will land on $(3, -1)$ after the transformation by K.

6. What happens to the unit vector $\vec{b} = (0,1)$ after the transformation given by matrix K.

$$K = \begin{bmatrix} 3 & -5 \\ -1 & 0 \end{bmatrix}$$

Solution:

Where $\vec{b} = (0,1)$ lands is given by the second column of the transformation matrix. So $\vec{b}$ will land on $(-5,0)$ after the transformation by K.

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PREIMAGE, IMAGE, AND THE KERNEL

1. Find the preimage A_1 of the subset B_1 under the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$.

$$B_1 = \left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \end{bmatrix} \right\}$$

$$T(\vec{x}) = \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution:

We're trying to find the preimage of B_1 under T, which we'll call $T^{-1}(B_1)$.

$$T^{-1}(B_1) = \{ \vec{x} \in \mathbb{R}^2 \mid T(\vec{x}) \in B_1 \}$$

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^2 \;\middle|\; \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} \right\}$$

In other words, we're just trying to find all the vectors $\vec{x}$ in $\mathbb{R}^2$ that satisfy either of these matrix equations. So let's rewrite both augmented matrices in reduced row-echelon form. We get

$$\begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$$\left[\begin{array}{cc|c} -2 & 4 & 0 \\ 3 & -1 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 3 & -1 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 5 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]$$

and

$$\begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} -2 & 4 & -1 \\ 3 & -1 & 4 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & \frac{1}{2} \\ 3 & -1 & 4 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & -2 & \frac{1}{2} \\ 0 & 5 & \frac{5}{2} \end{array}\right]$$

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$$\left[\begin{array}{ccc} 1 & -2 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

$$\left[\begin{array}{ccc} 1 & 0 & \frac{3}{2} \\ 0 & 1 & \frac{1}{2} \end{array}\right]$$

From the first augmented matrix, we get $x_1 = 0$ and $x_2 = 0$. And from the second augmented matrix we get $x_1 = 3/2$ and $x_2 = 1/2$. Therefore,

$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ in the subset B_1 under T

$\begin{bmatrix} \frac{3}{2} \\ \frac{1}{2} \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$ in the subset B_1 under T

2. Find the preimage A_1 of the subset B_1 under the transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$.

$$B_1 = \left\{ \begin{bmatrix} 1 \\ -4 \end{bmatrix}, \begin{bmatrix} -5 \\ 2 \end{bmatrix} \right\}$$

$$T(\vec{x}) = \begin{bmatrix} -1 & -3 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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Solution:

We're trying to find the preimage of B_1 under T, which we'll call $T^{-1}(B_1)$.

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \ \middle| \ T(\vec{x}) \in B_1 \right\}$$

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \ \middle| \ \begin{bmatrix} -1 & -3 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix} \text{ or } \begin{bmatrix} -1 & -3 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \end{bmatrix} \right\}$$

In other words, we're just trying to find all the vectors $\vec{x}$ in $\mathbb{R}^3$ that satisfy either of these matrix equations. So let's rewrite both augmented matrices in reduced row-echelon form. We get

$$\begin{bmatrix} -1 & -3 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & -3 & 2 & 1 \\ -2 & 0 & -2 & -4 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & -1 \\ -2 & 0 & -2 & -4 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & -1 \\ 0 & 6 & -6 & -6 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & -1 \\ 0 & 1 & -1 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -1 \end{array}\right]$$

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and

$$\begin{bmatrix} -1 & -3 & 2 \\ -2 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & -3 & 2 & -5 \\ -2 & 0 & -2 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ -2 & 0 & -2 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 6 & -6 & 12 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 5 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right]$$

The first augmented matrix gives a system of linear equations

$$x_1 + x_3 = 2$$

$$x_2 - x_3 = -1$$

that can be solved for the pivot variables in terms of the free variables.

$$x_1 = 2 - x_3$$

$$x_2 = -1 + x_3$$

The second augmented matrix gives a system of linear equations

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$$x_1 + x_3 = -1$$

$$x_2 - x_3 = 2$$

that can be solved for the pivot variables in terms of the free variables.

$$x_1 = -1 - x_3$$

$$x_2 = 2 + x_3$$

From the first augmented matrix, we get $x_1 = 2 - x_3$ and $x_2 = x_3 - 1$. And from the second augmented matrix we get $x_1 = -1 - x_3$ and $x_2 = 2 + x_3$. Therefore,

$\begin{bmatrix} 2 - x_3 \\ x_3 - 1 \\ x_3 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} 1 \\ -4 \end{bmatrix}$ in the subset B_1

under T

$\begin{bmatrix} -1 - x_3 \\ 2 + x_3 \\ x_3 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ in the subset B_1

under T

■ 3. Find the kernel of the transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$.

$$T(\vec{x}) = \begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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Solution:

The kernel of a transformation T is all of the vectors that result in the zero vector under the transformation T:

$$\text{Ker}(T) = \left\{ \vec{x} \in \mathbb{R}^2 \;\middle|\; T(\vec{x}) = \vec{O} \right\}$$

$$\begin{bmatrix} 3 & 9 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 3 & 9 & 0 \\ -1 & -3 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 3 & 0 \\ -1 & -3 & 0 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 0 & 0 \end{array}\right]$$

The augmented matrix gives the equation

$$x_1 + 3x_2 = 0$$

which can be solved for the pivot variable in terms of the free variable.

$$x_1 = -3x_2$$

or

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

Then the kernel of the transformation is

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$$\text{Ker}(T) = \text{Span}\left\{ \begin{bmatrix} -3 \\ 1 \end{bmatrix} \right\}$$

4. Find the kernel of the transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$.

$$T(\vec{x}) = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

The kernel of a transformation T is all of the vectors that result in the zero vector under the transformation T:

$$\text{Ker}(T) = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; T(\vec{x}) = \vec{O} \right\}$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 2 & 1 & -1 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 2 & 1 & -1 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right]$$

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The gives a system of linear equations

$$x_1 - x_3 = 0$$

$$x_2 + x_3 = 0$$

that can be solved for the pivot variables in terms of the free variable.

$$x_1 = x_3$$

$$x_2 = -x_3$$

or

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Then the kernel of the transformation is

$$\text{Ker}(T) = \text{Span}\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

5. Find the preimage A_1 of the subset B_1 under the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$.

$$B_1 = \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix} \right\}$$

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$$T(\vec{x}) = \begin{bmatrix} 1 & 1 & -2 \\ -3 & -4 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

We're trying to find the preimage of B_1 under T, which we'll call $T^{-1}(B_1)$.

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; T(\vec{x}) \in B_1 \right\}$$

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} 1 & 1 & -2 \\ -3 & -4 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \right.$$

$$\left. \text{or } \begin{bmatrix} 1 & 1 & -2 \\ -3 & -4 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix} \right\}$$

In other words, we're just trying to find all the vectors $\vec{x}$ in $\mathbb{R}^3$ that satisfy either of these matrix equations. So let's rewrite both augmented matrices in reduced row-echelon form. We get

$$\begin{bmatrix} 1 & 1 & -2 \\ -3 & -4 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ -3 & -4 & 0 & 1 \\ 0 & -2 & 1 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 0 & -1 & -6 & -5 \\ 0 & -2 & 1 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 0 & 1 & 6 & 5 \\ 0 & -2 & 1 & 3 \end{array}\right]$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & -7 \\ 0 & 1 & 6 & 5 \\ 0 & -2 & 1 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -8 & -7 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & 13 & 13 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -8 & -7 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 6 & 5 \\ 0 & 0 & 1 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

and

$$\begin{bmatrix} 1 & 1 & -2 \\ -3 & -4 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & 4 \\ -3 & -4 & 0 & -2 \\ 0 & -2 & 1 & 7 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & 4 \\ 0 & -1 & -6 & 10 \\ 0 & -2 & 1 & 7 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -2 & 4 \\ 0 & 1 & 6 & -10 \\ 0 & -2 & 1 & 7 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & 14 \\ 0 & 1 & 6 & -10 \\ 0 & -2 & 1 & 7 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -8 & 14 \\ 0 & 1 & 6 & -10 \\ 0 & 0 & 13 & -13 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -8 & 14 \\ 0 & 1 & 6 & -10 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 6 & -10 \\ 0 & 0 & 1 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -1 \end{array}\right]$$

From the first augmented matrix, we get $x_1 = 1$, $x_2 = -1$, and $x_3 = 1$. And from the second augmented matrix we get $x_1 = 6$, $x_2 = -4$, and $x_3 = -1$. Therefore,

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$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix}$ in the subset B_1 under T

$\begin{bmatrix} 6 \\ -4 \\ -1 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} 4 \\ -2 \\ 7 \end{bmatrix}$ in the subset B_1 under T

6. Find the preimage A_1 of the subset B_1 under the transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$.

$$B_1 = \left\{ \begin{bmatrix} -4 \\ 4 \\ -12 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix} \right\}$$

$$T(\vec{x}) = \begin{bmatrix} -2 & 4 & -6 \\ 1 & -3 & 0 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

We're trying to find the preimage of B_1 under T, which we'll call $T^{-1}(B_1)$.

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; T(\vec{x}) \in B_1 \right\}$$

$$T^{-1}(B_1) = \left\{ \vec{x} \in \mathbb{R}^3 \;\middle|\; \begin{bmatrix} -2 & 4 & -6 \\ 1 & -3 & 0 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ -12 \end{bmatrix} \right\}$$

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$$\text{or } \begin{bmatrix} -2 & 4 & -6 \\ 1 & -3 & 0 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix} \Bigg\}$$

In other words, we're just trying to find all the vectors $\vec{x}$ in $\mathbb{R}^3$ that satisfy either of these matrix equations. So let's rewrite both augmented matrices in reduced row-echelon form. We get

$$\begin{bmatrix} -2 & 4 & -6 \\ 1 & -3 & 0 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ -12 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 4 & -6 & | & -4 \\ 1 & -3 & 0 & | & 4 \\ -2 & 1 & -1 & | & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 1 & -3 & 0 & | & 4 \\ -2 & 1 & -1 & | & -12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 0 & -1 & -3 & | & 2 \\ -2 & 1 & -1 & | & -12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 0 & -1 & -3 & | & 2 \\ 0 & -3 & 5 & | & -8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -2 & 3 & | & 2 \\ 0 & 1 & 3 & | & -2 \\ 0 & -3 & 5 & | & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 9 & | & -2 \\ 0 & 1 & 3 & | & -2 \\ 0 & -3 & 5 & | & -8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 9 & | & -2 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 14 & | & -14 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 9 & | & -2 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 1 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 7 \\ 0 & 1 & 3 & | & -2 \\ 0 & 0 & 1 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 7 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & -1 \end{bmatrix}$$

and

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$$\begin{bmatrix} -2 & 4 & -6 \\ 1 & -3 & 0 \\ -2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -2 & 4 & -6 & 2 \\ 1 & -3 & 0 & -3 \\ -2 & 1 & -1 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 1 & -3 & 0 & -3 \\ -2 & 1 & -1 & 3 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & -1 & -3 & -2 \\ -2 & 1 & -1 & 3 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & -1 & -3 & -2 \\ 0 & -3 & 5 & 1 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 1 & 3 & 2 \\ 0 & -3 & 5 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 9 & 3 \\ 0 & 1 & 3 & 2 \\ 0 & -3 & 5 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 9 & 3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 14 & 7 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 9 & 3 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -\frac{3}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{array}\right]$$

From the first augmented matrix, we get $x_1 = 7$, $x_2 = 1$, and $x_3 = -1$. And from the second augmented matrix we get $x_1 = -3/2$, $x_2 = 1/2$, and $x_3 = 1/2$. Therefore,

$\begin{bmatrix} 7 \\ 1 \\ -1 \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} -4 \\ 4 \\ 12 \end{bmatrix}$ in the subset B_1 under T

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$\begin{bmatrix} -\frac{3}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$ in the pre-image A_1 would map to $\begin{bmatrix} 2 \\ -3 \\ 3 \end{bmatrix}$ in the subset B_1 under

T

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LINEAR TRANSFORMATIONS AS MATRIX-VECTOR PRODUCTS

■ 1. Use a matrix-vector product to reflect the square with vertices $(3, -1)$, $(1, -1)$, $(1,1)$, and $(3,1)$ over the y-axis. What are the vertices of the reflected square? Graph the resulting figure.

Solution:

If each point in the square is given by (x, y), a reflection over the y-axis means we'll take the x-coordinate of each point in the square and multiply it by -1. So after the reflection, each transformed point will be $(-x, y)$.

Therefore, if a position vector

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original square, then the position vector

$$\vec{v} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

represents the corresponding point in the transformed square. So a transformation T that expresses the reflection for any vector in $\mathbb{R}^2$ is

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

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Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-1\\0\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-0\\1\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

Which means we can actually rewrite the transformation T as

$$T\left(\begin{bmatrix}v_1\\v_2\end{bmatrix}\right) = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}v_1\\v_2\end{bmatrix}$$

Now that we've built the transformation matrix, we can apply it to each of the vertices of the square, $(3,-1)$, $(1,-1)$, $(1,1)$, and $(3,1)$.

$$T\left(\begin{bmatrix}3\\-1\end{bmatrix}\right) = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}3\\-1\end{bmatrix} = \begin{bmatrix}-1(3)+0(-1)\\0(3)+1(-1)\end{bmatrix} = \begin{bmatrix}-3\\-1\end{bmatrix}$$

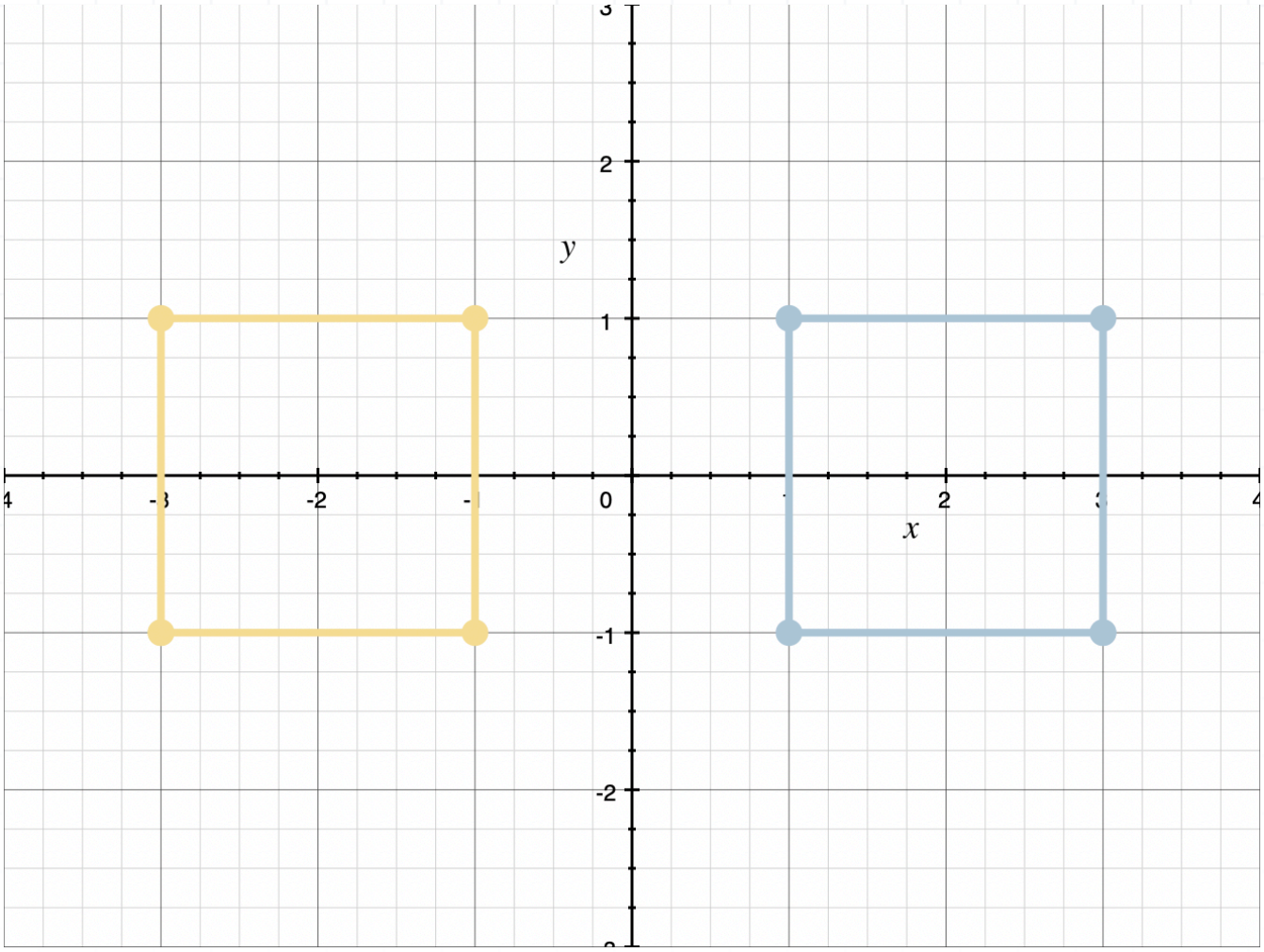
$$T\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}-1(1)+0(-1)\\0(1)+1(-1)\end{bmatrix} = \begin{bmatrix}-1\\-1\end{bmatrix}$$

$$T\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}-1(1)+0(1)\\0(1)+1(1)\end{bmatrix} = \begin{bmatrix}-1\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}3\\1\end{bmatrix}\right) = \begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}-1(3)+0(1)\\0(3)+1(1)\end{bmatrix} = \begin{bmatrix}-3\\1\end{bmatrix}$$

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A sketch of the original figure and its transformation is



■ 2. Use a matrix-vector product to transform the triangle with vertices $(-2, -3)$, $(4,2)$, and $(2, -5)$. The transformation T should include a reflection over the x-axis and horizontal stretch by a factor of 5. Graph the resulting figure.

Solution:

If each point in the triangle is given by (x, y), a reflection over the x-axis means we'll take the y-coordinate of each point in the triangle and multiply it by -1. So after the reflection, each transformed point will be $(x, -y)$.

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To stretch horizontally by a factor of 5, we'll need to multiply every x-value by 5. So after both the reflection and the horizontal stretch, each transformed point will be $(5x, -y)$.

Therefore, if a position vector

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original triangle, then the position vector

$$\vec{v} = \begin{bmatrix} 5v_1 \\ -v_2 \end{bmatrix}$$

represents the corresponding point in the transformed triangle. So a transformation T that expresses the reflection and the stretch for any vector in $\mathbb{R}^2$ is

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 5v_1 \\ -v_2 \end{bmatrix}$$

Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 5(1) \\ -0 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 5(0) \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

Which means we can actually rewrite the transformation T as

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$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Now that we've built the transformation matrix, we can apply it to each of the vertices of the triangle, $(-2, -3)$, $(4,2)$, and $(2, -5)$.

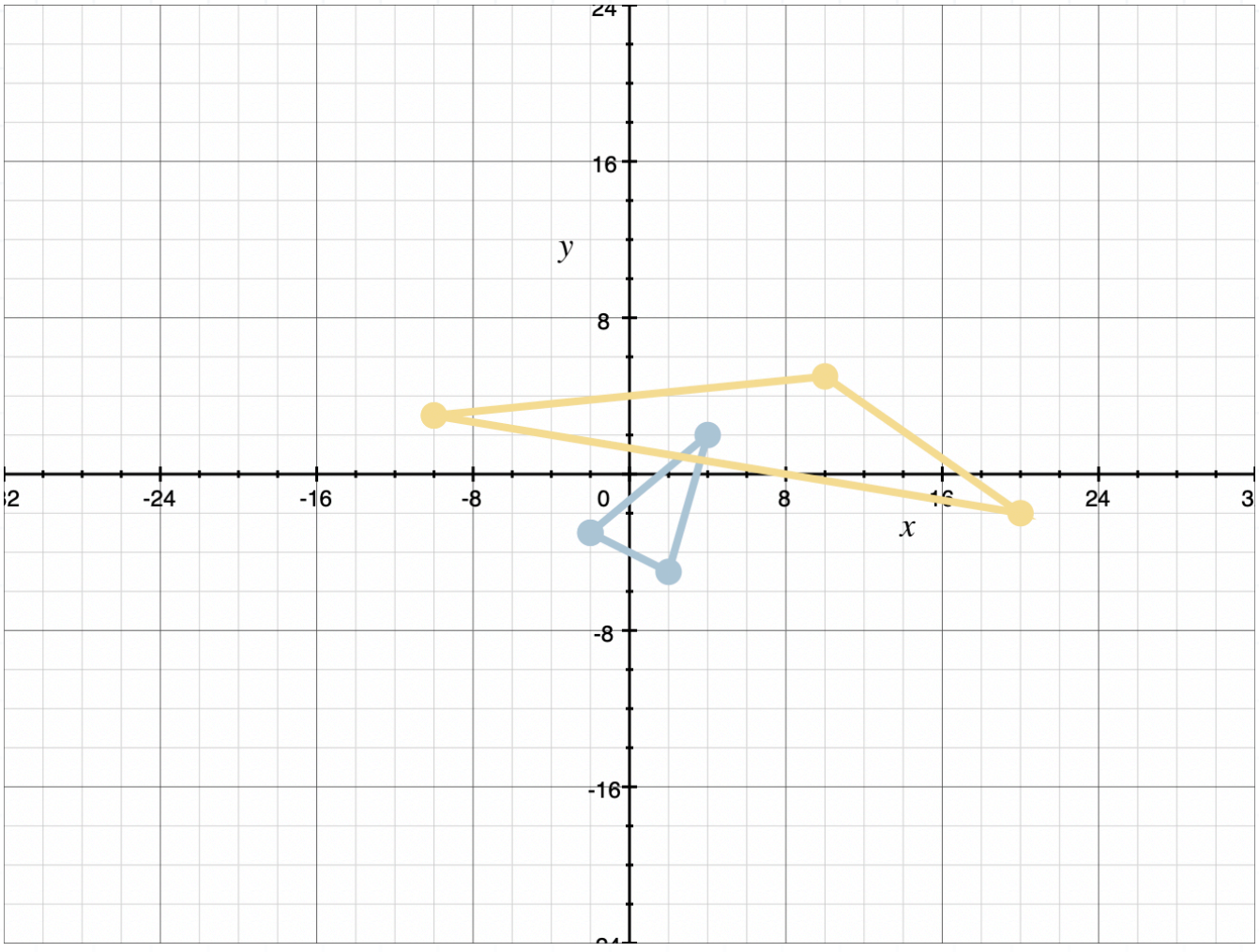
$$T\left(\begin{bmatrix} -2 \\ -3 \end{bmatrix}\right) = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -2 \\ -3 \end{bmatrix} = \begin{bmatrix} 5(-2) + 0(-3) \\ 0(-2) - 1(-3) \end{bmatrix} = \begin{bmatrix} -10 \\ 3 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 5(4) + 0(2) \\ 0(4) - 1(2) \end{bmatrix} = \begin{bmatrix} 20 \\ -2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 2 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} 5(2) + 0(-5) \\ 0(2) - 1(-5) \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

A sketch of the original figure and its transformation is

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3. Use a matrix-vector product to reflect the triangle with vertices $(3,3)$, $(1,-2)$, and $(-3,3)$ over the line $y = x$. What are the vertices of the reflected triangle? Graph the resulting figure.

Solution:

If each point in the triangle is given by (x, y), a reflection over the line $y = x$ means we'll take each point in the square and reverse x and y-values. So after the reflection, each transformed point will be (y, x).

Therefore, if a position vector

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$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original triangle, then a position vector

$$\vec{v} = \begin{bmatrix} v_2 \\ v_1 \end{bmatrix}$$

represents the corresponding point in the transformed triangle. So a transformation T that expresses the reflection for any vector in $\mathbb{R}^2$ is

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} v_2 \\ v_1 \end{bmatrix}$$

Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Which means we can actually rewrite the transformation T as

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Now that we've built the transformation matrix, we can apply it to each of the vertices of the triangle, $(3,3)$, $(1,-2)$, and $(-3,3)$.

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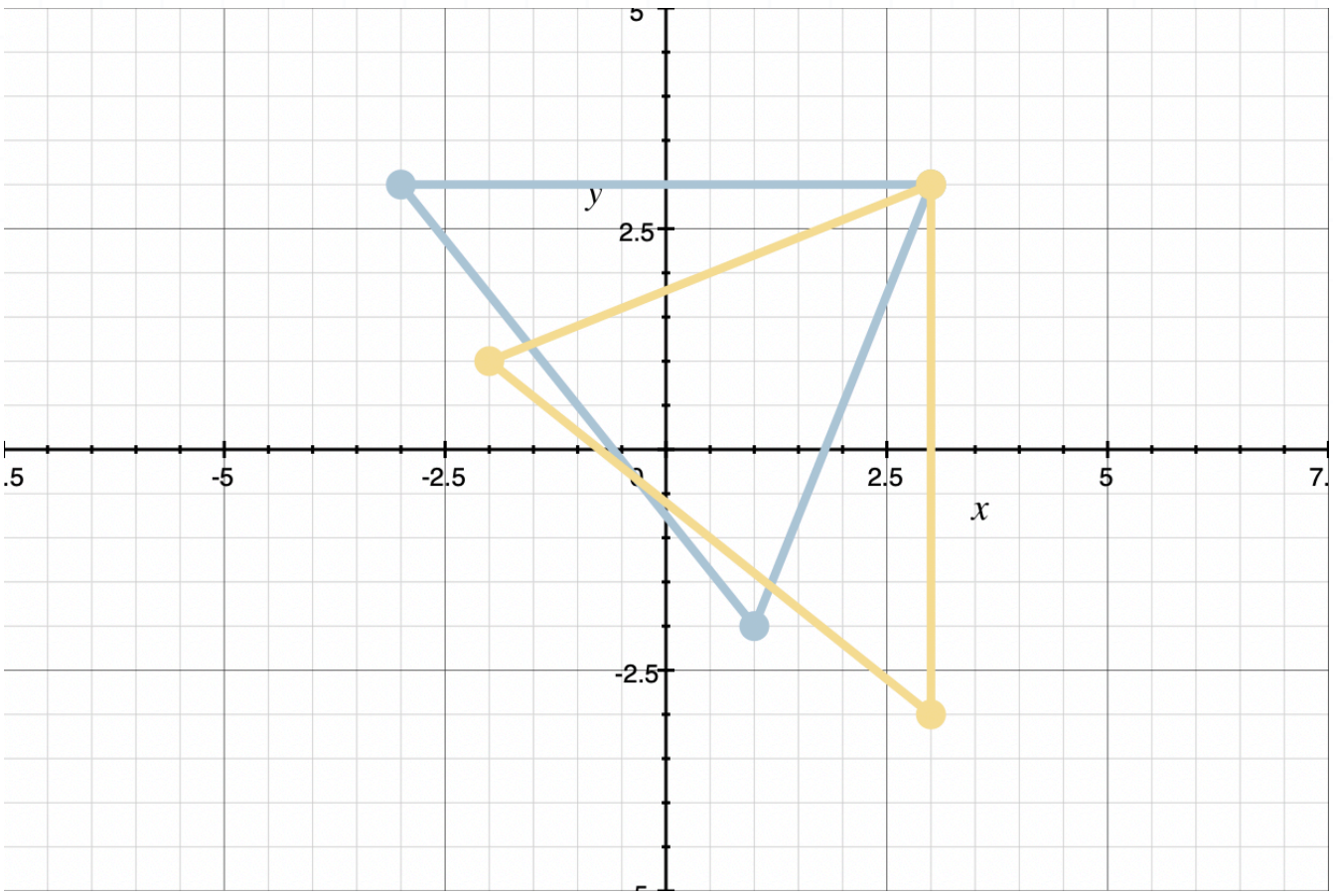


$$T\left(\begin{bmatrix}3\\3\end{bmatrix}\right)=\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}3\\3\end{bmatrix}=\begin{bmatrix}0(3)+1(3)\\1(3)+0(3)\end{bmatrix}=\begin{bmatrix}3\\3\end{bmatrix}$$

$$T\left(\begin{bmatrix}1\\-2\end{bmatrix}\right)=\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}1\\-2\end{bmatrix}=\begin{bmatrix}0(1)+1(-2)\\1(1)+0(-2)\end{bmatrix}=\begin{bmatrix}-2\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}-3\\3\end{bmatrix}\right)=\begin{bmatrix}0&1\\1&0\end{bmatrix}\begin{bmatrix}-3\\3\end{bmatrix}=\begin{bmatrix}0(-3)+1(3)\\1(-3)+0(3)\end{bmatrix}=\begin{bmatrix}3\\-3\end{bmatrix}$$

A sketch of the original figure and its transformation is



■ 4. Use a matrix-vector product to triple the width of the rectangle that has vertices $(-2,-1)$, $(-2,-5)$, $(5,-1)$, and $(5,-5)$, and then compress it vertically by a factor of 2. What are the vertices of the transformed rectangle? Graph the resulting figure.

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Solution:

Tripling the width of the rectangle means we're stretching it horizontally by a factor of 3.

If each point in the rectangle is given by (x, y), then tripling the width means we'll take the x-coordinate of each point in the rectangle and multiply it by 3. So after the stretch, each transformed point will be $(3x, y)$.

Therefore, if a position vector

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original rectangle, then a position vector

$$\vec{v} = \begin{bmatrix} 3v_1 \\ v_2 \end{bmatrix}$$

represents the corresponding point in the transformed rectangle. So a transformation T that expresses the stretch for any vector in $\mathbb{R}^2$ is

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 3v_1 \\ v_2 \end{bmatrix}$$

A vertical compression by a factor of 2 means we'll take the y-coordinate of each point in the rectangle and multiply it by $1/2$. So after the compression and stretch, each transformed point will be

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$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 3v_1 \\ \frac{1}{2}v_2 \end{bmatrix}$$

Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3(1) \\ \frac{1}{2}(0) \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3(0) \\ \frac{1}{2}(1) \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix}$$

Which means we can actually rewrite the transformation T as

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Now that we've built the transformation matrix, we can apply it to each of the vertices of the rectangle, $(-2, -1)$, $(-2, -5)$, $(5, -1)$, and $(5, -5)$.

$$T\left(\begin{bmatrix} -2 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3(-2) + 0(-1) \\ 0(-2) + \frac{1}{2}(-1) \end{bmatrix} = \begin{bmatrix} -6 \\ -\frac{1}{2} \end{bmatrix}$$

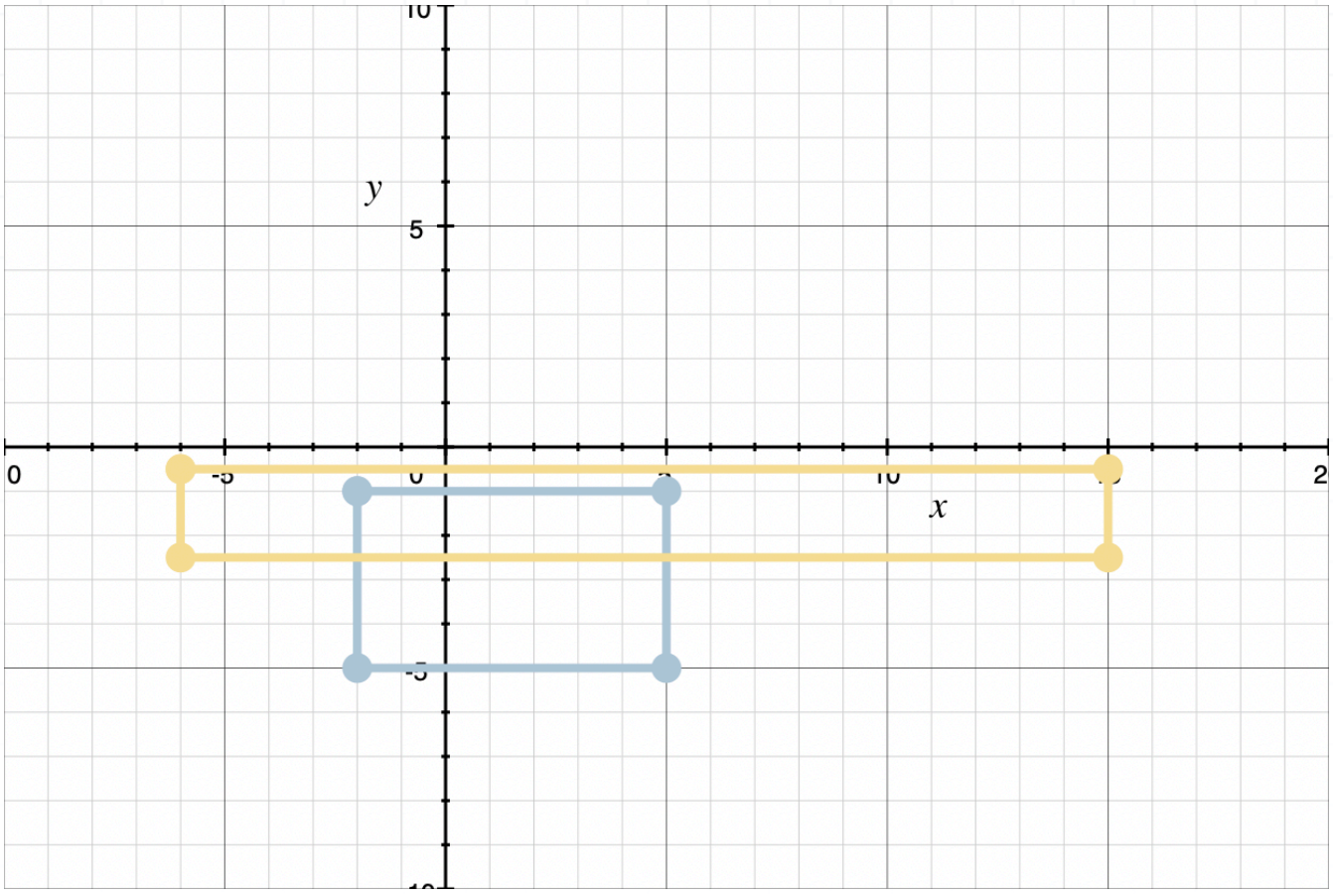
$$T\left(\begin{bmatrix} -2 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} -2 \\ -5 \end{bmatrix} = \begin{bmatrix} 3(-2) + 0(-5) \\ 0(-2) + \frac{1}{2}(-5) \end{bmatrix} = \begin{bmatrix} -6 \\ -\frac{5}{2} \end{bmatrix}$$

$$T\left(\begin{bmatrix} 5 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 3 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix} = \begin{bmatrix} 3(5) + 0(-1) \\ 0(5) + \frac{1}{2}(-1) \end{bmatrix} = \begin{bmatrix} 15 \\ -\frac{1}{2} \end{bmatrix}$$

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$$T\left(\begin{bmatrix}5\\-5\end{bmatrix}\right)=\begin{bmatrix}3&0\\0&\frac{1}{2}\end{bmatrix}\begin{bmatrix}5\\-5\end{bmatrix}=\begin{bmatrix}3(5)+0(-5)\\0(5)+\frac{1}{2}(-5)\end{bmatrix}=\begin{bmatrix}15\\-\frac{5}{2}\end{bmatrix}$$

A sketch of the original figure and its transformation is



■ 5. Use a matrix-vector product to reflect the parallelogram with vertices $(-3,1)$, $(0,4)$, $(7,4)$, and $(4,1)$ over the x-axis, and then over the y-axis. What are the vertices of the reflected parallelogram? Graph the resulting figure.

Solution:

If each point in the parallelogram is given by (x, y), a reflection over the x -axis means we'll take the y-coordinate of each point in the parallelogram

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and multiply it by -1. So after the reflection, each transformed point will be $(x, -y)$.

Therefore, if a position vector

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original parallelogram, then a position vector

$$\vec{v} = \begin{bmatrix} v_1 \\ -v_2 \end{bmatrix}$$

represents the corresponding point in the transformed parallelogram. So a transformation T that expresses the reflection for any vector in $\mathbb{R}^2$ is

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} v_1 \\ -v_2 \end{bmatrix}$$

A reflection over the y-axis means we'll take the x-coordinate of each point in the parallelogram and multiply it by -1. So after the two reflections, each transformed point will be

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -v_1 \\ -v_2 \end{bmatrix}$$

Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ -0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

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$$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right)=\begin{bmatrix}-0\\-1\end{bmatrix}=\begin{bmatrix}0\\-1\end{bmatrix}$$

Which means we can actually rewrite the transformation T as

$$T\left(\begin{bmatrix}v_1\\v_2\end{bmatrix}\right)=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}v_1\\v_2\end{bmatrix}$$

Now that we've built the transformation matrix, we can apply it to each of the vertices of the parallelogram, $(-3,1)$, $(0,4)$, $(7,4)$, and $(4,1)$.

$$T\left(\begin{bmatrix}-3\\1\end{bmatrix}\right)=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}-3\\1\end{bmatrix}=\begin{bmatrix}-1(-3)+0(1)\\0(-3)-1(1)\end{bmatrix}=\begin{bmatrix}3\\-1\end{bmatrix}$$

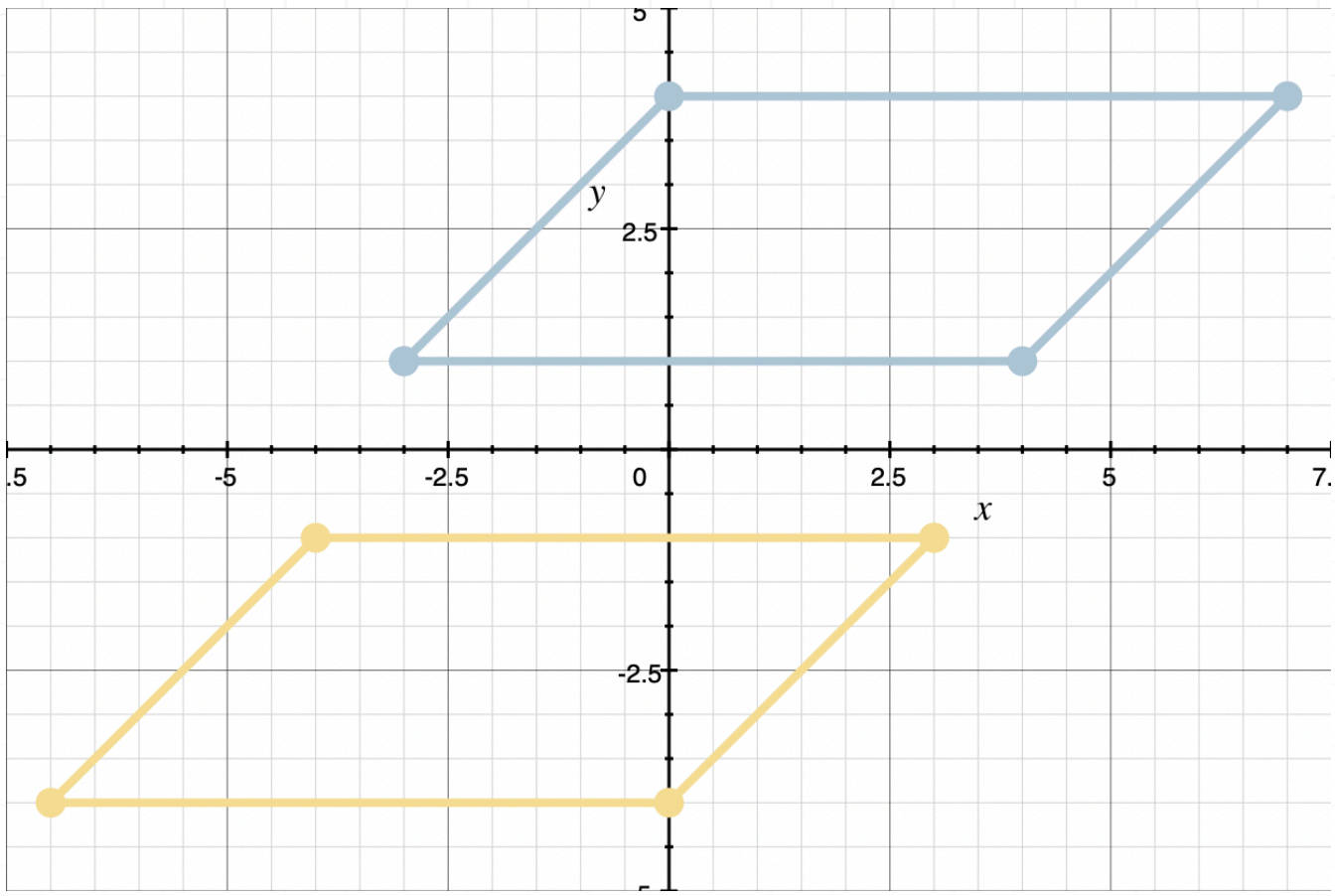
$$T\left(\begin{bmatrix}0\\4\end{bmatrix}\right)=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}0\\4\end{bmatrix}=\begin{bmatrix}-1(0)+0(4)\\0(0)-1(4)\end{bmatrix}=\begin{bmatrix}0\\-4\end{bmatrix}$$

$$T\left(\begin{bmatrix}7\\4\end{bmatrix}\right)=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}7\\4\end{bmatrix}=\begin{bmatrix}-1(7)+0(4)\\0(7)-1(4)\end{bmatrix}=\begin{bmatrix}-7\\-4\end{bmatrix}$$

$$T\left(\begin{bmatrix}4\\1\end{bmatrix}\right)=\begin{bmatrix}-1&0\\0&-1\end{bmatrix}\begin{bmatrix}4\\1\end{bmatrix}=\begin{bmatrix}-1(4)+0(1)\\0(4)-1(1)\end{bmatrix}=\begin{bmatrix}-4\\-1\end{bmatrix}$$

A sketch of the original figure and its transformation is

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■ 6. Use a matrix-vector product to reflect the triangle with vertices $(2,3)$, $(-5,-4)$, and $(-4,5)$ over the y-axis, and then stretch it horizontally by a factor of 4. What are the vertices of the transformed triangle? Graph the resulting figure.

Solution:

If each point in the triangle is given by (x, y), a reflection over the y-axis means we'll take the x-coordinate of each point in the triangle and multiply it by -1. So after the reflection, each transformed point will be $(-x, y)$.

Therefore, if a position vector

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$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

represents a point in the original triangle, then a position vector

$$\vec{v} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix}$$

represents the corresponding point in the transformed triangle.

A horizontal stretch by a factor of 4 means we'll multiply each x-value by 4. So after both transformations, each transformed point will be

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -4v_1 \\ v_2 \end{bmatrix}$$

Because we're transforming *from* $\mathbb{R}^2$, we can use T to transform each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -4(1) \\ 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -4(0) \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Which means we can actually rewrite the transformation T as

$$T\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

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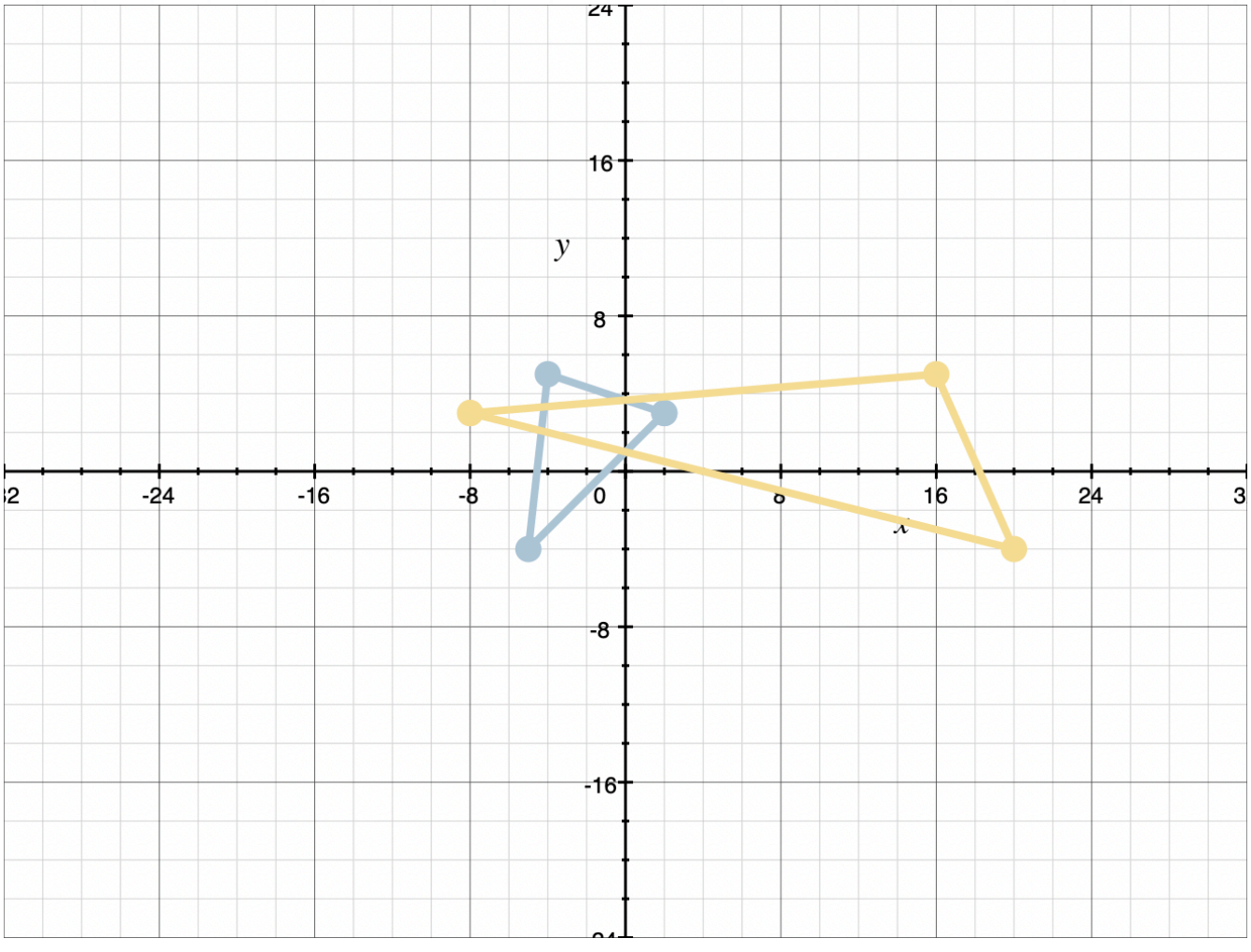
Now that we've built the transformation matrix, we can apply it to each of the vertices of the triangle, $(2,3)$, $(-5,-4)$, and $(-4,5)$.

$$T\left(\begin{bmatrix}2\\3\end{bmatrix}\right) = \begin{bmatrix}-4 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}-4(2)+0(3)\\0(2)+1(3)\end{bmatrix} = \begin{bmatrix}-8\\3\end{bmatrix}$$

$$T\left(\begin{bmatrix}-5\\-4\end{bmatrix}\right) = \begin{bmatrix}-4 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}-5\\-4\end{bmatrix} = \begin{bmatrix}-4(-5)+0(-4)\\0(-5)+1(-4)\end{bmatrix} = \begin{bmatrix}20\\-4\end{bmatrix}$$

$$T\left(\begin{bmatrix}-4\\5\end{bmatrix}\right) = \begin{bmatrix}-4 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}-4\\5\end{bmatrix} = \begin{bmatrix}-4(-4)+0(5)\\0(-4)+1(5)\end{bmatrix} = \begin{bmatrix}16\\5\end{bmatrix}$$

A sketch of the original figure and its transformation is



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LINEAR TRANSFORMATIONS AS ROTATIONS

1. Find the rotation of $\vec{x} = (2, -4)$ by an angle of $\theta = 120°$.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^2$ by $120°$ is

$$\text{Rot}_{120°}(\vec{x}) = \begin{bmatrix} \cos(120°) & -\sin(120°) \\ \sin(120°) & \cos(120°) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Simplify the rotation matrix. We can get the sine and cosine values at $\theta = 120°$ from the unit circle, or from a calculator.

$$\begin{bmatrix} \cos(120°) & -\sin(120°) \\ \sin(120°) & \cos(120°) \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\left(\frac{\sqrt{3}}{2}\right) \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}$$

So the transformation for a $120°$ rotation is

$$\text{Rot}_{120°}(\vec{x}) = \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Apply this specific rotation matrix to $\vec{x} = (2, -4)$.

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$$\text{Rot}_{120°}\left(\begin{bmatrix}2\\-4\end{bmatrix}\right) = \begin{bmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}\begin{bmatrix}2\\-4\end{bmatrix}$$

$$\text{Rot}_{120°}\left(\begin{bmatrix}2\\-4\end{bmatrix}\right) = \begin{bmatrix}-\frac{1}{2}(2) - \frac{\sqrt{3}}{2}(-4)\\ \frac{\sqrt{3}}{2}(2) - \frac{1}{2}(-4)\end{bmatrix}$$

$$\text{Rot}_{120°}\left(\begin{bmatrix}2\\-4\end{bmatrix}\right) = \begin{bmatrix}-1 + 2\sqrt{3}\\ \sqrt{3} + 2\end{bmatrix}$$

$$\text{Rot}_{120°}\left(\begin{bmatrix}2\\-4\end{bmatrix}\right) = \begin{bmatrix}-1 + 2\sqrt{3}\\ 2 + \sqrt{3}\end{bmatrix}$$

■ 2. Find the rotation of $\vec{x} = (1, -5)$ by an angle of $\theta = 60°$.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^2$ by $60°$ is

$$\text{Rot}_{60°}(\vec{x}) = \begin{bmatrix}\cos(60°) & -\sin(60°)\\ \sin(60°) & \cos(60°)\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}$$

Simplify the rotation matrix. We can get the sine and cosine values at $\theta = 60°$ from the unit circle, or from a calculator.

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$$\begin{bmatrix} \cos(60°) & -\sin(60°) \\ \sin(60°) & \cos(60°) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\left(\frac{\sqrt{3}}{2}\right) \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

So the transformation for a $60°$ rotation is

$$\text{Rot}_{60°}(\vec{x}) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Apply this specific rotation matrix to $\vec{x} = (1, -5)$.

$$\text{Rot}_{60°}\left(\begin{bmatrix} 1 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ -5 \end{bmatrix}$$

$$\text{Rot}_{60°}\left(\begin{bmatrix} 1 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2}(1) - \frac{\sqrt{3}}{2}(-5) \\ \frac{\sqrt{3}}{2}(1) + \frac{1}{2}(-5) \end{bmatrix}$$

$$\text{Rot}_{60°}\left(\begin{bmatrix} 1 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} + \frac{5\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} - \frac{5}{2} \end{bmatrix}$$

$$\text{Rot}_{60°}\left(\begin{bmatrix} 1 \\ -5 \end{bmatrix}\right) = \begin{bmatrix} \frac{1 + 5\sqrt{3}}{2} \\ \frac{-5 + \sqrt{3}}{2} \end{bmatrix}$$

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3. Find the rotation of $\vec{x} = (-7,4)$ by an angle of $\theta = 180°$.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^2$ by $180°$ is

$$\text{Rot}_{180°}(\vec{x}) = \begin{bmatrix} \cos(180°) & -\sin(180°) \\ \sin(180°) & \cos(180°) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Simplify the rotation matrix. We can get the sine and cosine values at $\theta = 180°$ from the unit circle, or from a calculator.

$$\begin{bmatrix} \cos(180°) & -\sin(180°) \\ \sin(180°) & \cos(180°) \end{bmatrix} = \begin{bmatrix} -1 & -0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

So the transformation for a $180°$ rotation is

$$\text{Rot}_{180°}(\vec{x}) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Apply this specific rotation matrix to $\vec{x} = (-7,4)$.

$$\text{Rot}_{180°}\left(\begin{bmatrix} -7 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -7 \\ 4 \end{bmatrix}$$

$$\text{Rot}_{180°}\left(\begin{bmatrix} -7 \\ 4 \end{bmatrix} \right) = \begin{bmatrix} -1(-7) + 0(4) \\ 0(-7) - 1(4) \end{bmatrix}$$

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$$\text{Rot}_{180^\circ}\left(\begin{bmatrix}-7\\4\end{bmatrix}\right)=\begin{bmatrix}7+0\\0-4\end{bmatrix}$$

$$\text{Rot}_{180^\circ}\left(\begin{bmatrix}-7\\4\end{bmatrix}\right)=\begin{bmatrix}7\\-4\end{bmatrix}$$

■ 4. Find the rotation of $\vec{x}=(-4,1,3)$ by an angle of $\theta=90^\circ$ about the x-axis.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^3$ by 90° around the x-axis is

$$\text{Rot}_{90^\circ \text{ around } x}=\begin{bmatrix}1&0&0\\0&\cos(90^\circ)&-\sin(90^\circ)\\0&\sin(90^\circ)&\cos(90^\circ)\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$$

Simplify the rotation matrix. We can get the sine and cosine values at $\theta=90^\circ$ from the unit circle, or from a calculator.

$$\begin{bmatrix}1&0&0\\0&\cos(90^\circ)&-\sin(90^\circ)\\0&\sin(90^\circ)&\cos(90^\circ)\end{bmatrix}=\begin{bmatrix}1&0&0\\0&0&-1\\0&1&0\end{bmatrix}$$

So the transformation for a 90° rotation around the x-axis is

$$\text{Rot}_{90^\circ \text{ around } x}(\vec{x})=\begin{bmatrix}1&0&0\\0&0&-1\\0&1&0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$$

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Apply this specific rotation matrix to $\vec{x} = (-4,1,3)$.

$$\text{Rot}_{90^\circ \text{ around } x}\left(\begin{bmatrix}-4\\1\\3\end{bmatrix}\right) = \begin{bmatrix}1 & 0 & 0\\0 & 0 & -1\\0 & 1 & 0\end{bmatrix}\begin{bmatrix}-4\\1\\3\end{bmatrix}$$

$$\text{Rot}_{90^\circ \text{ around } x}\left(\begin{bmatrix}-4\\1\\3\end{bmatrix}\right) = \begin{bmatrix}1(-4)+0(1)+0(3)\\0(-4)+0(1)-1(3)\\0(-4)+1(1)+0(3)\end{bmatrix}$$

$$\text{Rot}_{90^\circ \text{ around } x}\left(\begin{bmatrix}-4\\1\\3\end{bmatrix}\right) = \begin{bmatrix}-4+0+0\\0+0-3\\0+1+0\end{bmatrix}$$

$$\text{Rot}_{90^\circ \text{ around } x}\left(\begin{bmatrix}-4\\1\\3\end{bmatrix}\right) = \begin{bmatrix}-4\\-3\\1\end{bmatrix}$$

■ 5. Find the rotation of $\vec{x} = (-2/\sqrt{2},2,0)$ by an angle of $\theta = 315^\circ$ about the y -axis.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^3$ by 315° around the y-axis is

$$\text{Rot}_{315^\circ \text{ around } y} = \begin{bmatrix}\cos(315^\circ) & 0 & \sin(315^\circ)\\0 & 1 & 0\\-\sin(315^\circ) & 0 & \cos(315^\circ)\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$$

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Simplify the rotation matrix. We can get the sine and cosine values at $\theta = 315°$ from the unit circle, or from a calculator.

$$\begin{bmatrix} \cos(315°) & 0 & \sin(315°) \\ 0 & 1 & 0 \\ -\sin(315°) & 0 & \cos(315°) \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ -\left(-\frac{\sqrt{2}}{2}\right) & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$$

So the transformation for a $315°$ rotation around the y-axis is

$$\text{Rot}_{315° \text{ around } y}(\vec{x}) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Apply this specific rotation matrix to $\vec{x} = (-2/\sqrt{2}, 2, 0)$.

$$\text{Rot}_{315° \text{ around } y}\left(\begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 2 \\ 0 \end{bmatrix}$$

$$\text{Rot}_{315° \text{ around } y}\left(\begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 2 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} \frac{\sqrt{2}}{2}\left(-\frac{2}{\sqrt{2}}\right) + 0(2) - \frac{\sqrt{2}}{2}(0) \\ 0\left(-\frac{2}{\sqrt{2}}\right) + 1(2) + 0(0) \\ \frac{\sqrt{2}}{2}\left(-\frac{2}{\sqrt{2}}\right) + 0(2) + \frac{\sqrt{2}}{2}(0) \end{bmatrix}$$

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$$\text{Rot}_{315^\circ \text{ around } y}\left(\begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 2 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1+0+0 \\ 0+2+0 \\ -1+0+0 \end{bmatrix}$$

$$\text{Rot}_{315^\circ \text{ around } y}\left(\begin{bmatrix} -\frac{2}{\sqrt{2}} \\ 2 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$$

■ 6. Find the rotation of $\vec{x} = (-2,0,3)$ by an angle of $\theta = 150^\circ$ about the z -axis.

Solution:

The transformation to rotate any vector $\vec{x}$ in $\mathbb{R}^3$ by 150° around the z-axis is

$$\text{Rot}_{150^\circ \text{ around } z} = \begin{bmatrix} \cos(150^\circ) & -\sin(150^\circ) & 0 \\ \sin(150^\circ) & \cos(150^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Simplify the rotation matrix. We can get the sine and cosine values at $\theta = 150^\circ$ from the unit circle, or from a calculator.

$$\begin{bmatrix} \cos(150^\circ) & -\sin(150^\circ) & 0 \\ \sin(150^\circ) & \cos(150^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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So the transformation for a $150°$ rotation around the z-axis is

$$\text{Rot}_{150° \text{ around } z}(\vec{x}) = \begin{bmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Apply this specific rotation matrix to $\vec{x} = (-2,0,3)$.

$$\text{Rot}_{150° \text{ around } z}\left(\begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

$$\text{Rot}_{150° \text{ around } z}\left(\begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} -\frac{\sqrt{3}}{2}(-2) - \frac{1}{2}(0) + 0(3) \\ \frac{1}{2}(-2) - \frac{\sqrt{3}}{2}(0) + 0(3) \\ 0(-2) + 0(0) + 1(3) \end{bmatrix}$$

$$\text{Rot}_{150° \text{ around } z}\left(\begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} \sqrt{3} + 0 + 0 \\ -1 + 0 + 0 \\ 0 + 0 + 3 \end{bmatrix}$$

$$\text{Rot}_{150° \text{ around } z}\left(\begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} \sqrt{3} \\ -1 \\ 3 \end{bmatrix}$$

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ADDING AND SCALING LINEAR TRANSFORMATIONS

1. Find the product of a scalar $c = 5$ and the transformation $T(\vec{x})$.

$$T(\vec{x}) = \begin{bmatrix} 0 & -4 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution:

The transformation T is given as a matrix-vector product. If we call the transformation matrix B, then multiplying the transformation by the scalar $c = 5$ means we multiply B by c.

$$cT(\vec{x}) = 5 \begin{bmatrix} 0 & -4 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

First find cB.

$$cB = 5 \begin{bmatrix} 0 & -4 \\ -3 & 1 \end{bmatrix}$$

$$cB = \begin{bmatrix} 5(0) & 5(-4) \\ 5(-3) & 5(1) \end{bmatrix}$$

$$cB = \begin{bmatrix} 0 & -20 \\ -15 & 5 \end{bmatrix}$$

So the scaled transformation would be

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$$cT(\vec{x}) = \begin{bmatrix} 0 & -20 \\ -15 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

2. Find the product of a scalar $c = -2$ and the transformation $T(\vec{x})$.

$$T(\vec{x}) = \begin{bmatrix} -1 & 0 & 4 \\ 3 & -5 & 7 \\ -2 & -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

The transformation T is given as a matrix-vector product. If we call the transformation matrix A, then multiplying the transformation by the scalar $c = -2$ means we multiply A by c.

$$cT(\vec{x}) = -2 \begin{bmatrix} -1 & 0 & 4 \\ 3 & -5 & 7 \\ -2 & -4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

First find cA.

$$cA = -2 \begin{bmatrix} -1 & 0 & 4 \\ 3 & -5 & 7 \\ -2 & -4 & 1 \end{bmatrix}$$

$$cA = \begin{bmatrix} -2(-1) & -2(0) & -2(4) \\ -2(3) & -2(-5) & -2(7) \\ -2(-2) & -2(-4) & -2(1) \end{bmatrix}$$

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$$cA = \begin{bmatrix} 2 & 0 & -8 \\ -6 & 10 & -14 \\ 4 & 8 & -2 \end{bmatrix}$$

So the scaled transformation would be

$$cT(\vec{x}) = \begin{bmatrix} 2 & 0 & -8 \\ -6 & 10 & -14 \\ 4 & 8 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

3. Find the sum of the transformations $S(\vec{x})$ and $T(\vec{x})$.

$$S(\vec{x}) = \begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -2 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution:

The sum of the transformations is the sum of the matrices A and B given in their matrix vector products,

$$S(\vec{x}) = \begin{bmatrix} -4 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -2 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The sum of the matrices is

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$$A+B=\begin{bmatrix}-4 & 3\\ 2 & 1\end{bmatrix}+\begin{bmatrix}-2 & 4\\ 0 & -1\end{bmatrix}$$

$$A+B=\begin{bmatrix}-4-2 & 3+4\\ 2+0 & 1-1\end{bmatrix}$$

$$A+B=\begin{bmatrix}-6 & 7\\ 2 & 0\end{bmatrix}$$

So the sum of the transformations would be

$$S(\vec{x})+T(\vec{x})=\begin{bmatrix}-6 & 7\\ 2 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

4. Find the sum of the transformations $S(\vec{x})$ and $T(\vec{x})$.

$$S(\vec{x})=\begin{bmatrix}0 & -4 & 1\\ 1 & -1 & 3\\ 2 & -1 & 4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

$$T(\vec{x})=\begin{bmatrix}5 & -3 & 3\\ 2 & 0 & -1\\ 1 & -4 & -5\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Solution:

The sum of the transformations is the sum of the matrices A and B given in their matrix vector products,

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$$S(\vec{x}) = \begin{bmatrix} 0 & -4 & 1 \\ 1 & -1 & 3 \\ 2 & -1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} 5 & -3 & 3 \\ 2 & 0 & -1 \\ 1 & -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The sum of the matrices is

$$A + B = \begin{bmatrix} 0 & -4 & 1 \\ 1 & -1 & 3 \\ 2 & -1 & 4 \end{bmatrix} + \begin{bmatrix} 5 & -3 & 3 \\ 2 & 0 & -1 \\ 1 & -4 & -5 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 0+5 & -4-3 & 1+3 \\ 1+2 & -1+0 & 3-1 \\ 2+1 & -1-4 & 4-5 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 5 & -7 & 4 \\ 3 & -1 & 2 \\ 3 & -5 & -1 \end{bmatrix}$$

So the sum of the transformations would be

$$S(\vec{x}) + T(\vec{x}) = \begin{bmatrix} 5 & -7 & 4 \\ 3 & -1 & 2 \\ 3 & -5 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

■ 5. Find the sum of the transformation $S(\vec{x})$ and the product of a scalar $c = -1/2$ and $T(\vec{x})$.

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$$S(\vec{x}) = \begin{bmatrix} -5 & 0 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -4 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution:

First find the product of a scalar $c = -1/2$ and the transformation $T(\vec{x})$.

The transformation T is given as a matrix-vector product. If we call the transformation matrix B, then multiplying the transformation by the scalar $c = -1/2$ means we multiply B by c.

$$cT(\vec{x}) = -\frac{1}{2} \begin{bmatrix} -4 & 2 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

First find cB.

$$cB = -\frac{1}{2} \begin{bmatrix} -4 & 2 \\ -2 & 0 \end{bmatrix}$$

$$cB = \begin{bmatrix} -\frac{1}{2}(-4) & -\frac{1}{2}(2) \\ -\frac{1}{2}(-2) & -\frac{1}{2}(0) \end{bmatrix}$$

$$cB = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

So the scaled transformation would be

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$$cT(\vec{x}) = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Now find the sum of the transformations $S(\vec{x})$ and $cT(\vec{x})$.

The sum of the transformations is the sum of the matrices A and $C = cB$ given in their matrix vector products,

$$S(\vec{x}) = \begin{bmatrix} -5 & 0 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$cT(\vec{x}) = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The sum of the matrices is

$$A + C = \begin{bmatrix} -5 & 0 \\ -2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A + C = \begin{bmatrix} -5+2 & 0-1 \\ -2+1 & -1+0 \end{bmatrix}$$

$$A + C = \begin{bmatrix} -3 & -1 \\ -1 & -1 \end{bmatrix}$$

So the sum of the transformations would be

$$S(\vec{x}) + cT(\vec{x}) = \begin{bmatrix} -3 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

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■ 6. Find the product of $c = 1/3$ with the sum of the transformations $S(\vec{x})$ and $T(\vec{x})$.

$$S(\vec{x}) = \begin{bmatrix} -5 & 4 & -3 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -1 & 2 & 0 \\ 6 & -4 & 5 \\ 9 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Solution:

First find the sum of the transformations $S(\vec{x})$ and $cT(\vec{x})$.

The sum of the transformations is the sum of the matrices A and B given in their matrix vector products,

$$S(\vec{x}) = \begin{bmatrix} -5 & 4 & -3 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -1 & 2 & 0 \\ 6 & -4 & 5 \\ 9 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

The sum of the matrices is

$$A + B = \begin{bmatrix} -5 & 4 & -3 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{bmatrix} + \begin{bmatrix} -1 & 2 & 0 \\ 6 & -4 & 5 \\ 9 & 1 & 3 \end{bmatrix}$$

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$$A+B=\begin{bmatrix}-5-1 & 4+2 & -3+0\\ 0+6 & 1-4 & -2+5\\ 0+9 & -1+1 & 3+3\end{bmatrix}$$

$$A+B=\begin{bmatrix}-6 & 6 & -3\\ 6 & -3 & 3\\ 9 & 0 & 6\end{bmatrix}$$

So the sum of the transformations would be

$$S(\vec{x})+T(\vec{x})=\begin{bmatrix}-6 & 6 & -3\\ 6 & -3 & 3\\ 9 & 0 & 6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Now find the product of a scalar $c=1/3$ and the sum of the transformations $S(\vec{x})+T(\vec{x})$.

If we call the matrix that's in the sum of the transformation $S(\vec{x})+T(\vec{x})$ the matrix $C=A+B$, then multiplying the transformation by the scalar $c=1/3$ gives

$$c(S(\vec{x})+T(\vec{x}))=\frac{1}{3}\begin{bmatrix}-6 & 6 & -3\\ 6 & -3 & 3\\ 9 & 0 & 6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

First find cC.

$$cC=\frac{1}{3}\begin{bmatrix}-6 & 6 & -3\\ 6 & -3 & 3\\ 9 & 0 & 6\end{bmatrix}$$

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$$cC = \begin{bmatrix} \frac{1}{3}(-6) & \frac{1}{3}(6) & \frac{1}{3}(-3) \\ \frac{1}{3}(6) & \frac{1}{3}(-3) & \frac{1}{3}(3) \\ \frac{1}{3}(9) & \frac{1}{3}(0) & \frac{1}{3}(6) \end{bmatrix}$$

$$cC = \begin{bmatrix} -2 & 2 & -1 \\ 2 & -1 & 1 \\ 3 & 0 & 2 \end{bmatrix}$$

So the scaled transformation would be

$$c(S(\vec{x}) + T(\vec{x})) = \begin{bmatrix} -2 & 2 & -1 \\ 2 & -1 & 1 \\ 3 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

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PROJECTIONS AS LINEAR TRANSFORMATIONS

■ 1. Find the projection of $\vec{v}$ onto L.

$$L = \left\{ c \begin{bmatrix} 4 \\ 2 \end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

Solution:

The line L is given as all the scaled versions of the vector $\vec{x} = (4,2)$. Then the projection of $\vec{v}$ onto L is given by

$$\text{Proj}_L(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \right) \vec{x}$$

$$\text{Proj}_L(\vec{v}) = \frac{\begin{bmatrix} 1 & 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}}{\begin{bmatrix} 4 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{1(4) + 4(2)}{4(4) + 2(2)} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\text{Proj}_L(\vec{v}) = \frac{12}{20} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \frac{3}{5} \cdot \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{12}{5} \\ \frac{6}{5} \end{bmatrix} = \begin{bmatrix} 2.4 \\ 1.2 \end{bmatrix}$$

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■ 2. Find the projection of $\vec{v}$ onto M.

$$M = \left\{ c \begin{bmatrix} -4 \\ 3 \end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix} 0 \\ -5 \end{bmatrix}$$

Solution:

The line M is given as all the scaled versions of the vector $\vec{x} = (-4,3)$. Then the projection of $\vec{v}$ onto M is given by

$$\text{Proj}_M(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \right) \vec{x}$$

$$\text{Proj}_M(\vec{v}) = \frac{\begin{bmatrix} 0 & -5 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix}}{\begin{bmatrix} -4 & 3 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix}} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix} = \frac{0(-4) - 5(3)}{-4(-4) + 3(3)} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix}$$

$$\text{Proj}_M(\vec{v}) = \frac{-15}{25} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix} = -\frac{3}{5} \cdot \begin{bmatrix} -4 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{12}{5} \\ -\frac{9}{5} \end{bmatrix} = \begin{bmatrix} 2.4 \\ -1.8 \end{bmatrix}$$

■ 3. Find the projection of $\vec{v}$ onto L and the vector complement of $\vec{v}$ orthogonal to L.

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$$L = \left\{ c \begin{bmatrix} -3 \\ 1 \end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix} -4 \\ 0 \end{bmatrix}$$

Solution:

The line L is given as all the scaled versions of the vector $\vec{x} = (-3,1)$. Then the projection of $\vec{v}$ onto L is given by

$$\text{Proj}_L(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \right) \vec{x}$$

$$\text{Proj}_L(\vec{v}) = \frac{\begin{bmatrix} -4 & 0 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix}}{\begin{bmatrix} -3 & 1 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix}} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \frac{-4(-3) + 0(1)}{-3(-3) + 1(1)} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

$$\text{Proj}_L(\vec{v}) = \frac{12}{10} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \frac{6}{5} \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{18}{5} \\ \frac{6}{5} \end{bmatrix} = \begin{bmatrix} -3.6 \\ 1.2 \end{bmatrix}$$

The vector complement of $\vec{v}$ orthogonal to L is the vector $\vec{v} - \text{Proj}_L(\vec{v})$.

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} -4 \\ 0 \end{bmatrix} - \begin{bmatrix} -3.6 \\ 1.2 \end{bmatrix}$$

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} -4 - (-3.6) \\ 0 - 1.2 \end{bmatrix}$$

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$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} -0.4 \\ -1.2 \end{bmatrix}$$

■ 4. Find the projection of $\vec{v}$ onto L and the vector complement of $\vec{v}$ orthogonal to L.

$$L = \left\{ c \begin{bmatrix} -2 \\ 0 \end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix} 3 \\ -7 \end{bmatrix}$$

Solution:

The line L is given as all the scaled versions of the vector $\vec{x} = (-2,0)$. Then the projection of $\vec{v}$ onto L is given by

$$\text{Proj}_L(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \right) \vec{x}$$

$$\text{Proj}_L(\vec{v}) = \frac{\begin{bmatrix} 3 & -7 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix}}{\begin{bmatrix} -2 & 0 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix}} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \frac{3(-2) - 7(0)}{-2(-2) + 0(0)} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix}$$

$$\text{Proj}_L(\vec{v}) = \frac{-6}{4} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix} = -\frac{3}{2} \cdot \begin{bmatrix} -2 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

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The vector complement of $\vec{v}$ orthogonal to L is the vector $\vec{v} - \text{Proj}_L(\vec{v})$.

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} 3 \\ -7 \end{bmatrix} - \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} 3-3 \\ -7-0 \end{bmatrix}$$

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix} 0 \\ -7 \end{bmatrix}$$

5. Find the projection of $\vec{v}$ onto L.

$$L = \left\{ c \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

Solution:

The line L is given as all the scaled versions of the vector $\vec{x} = (-2,1,3)$. Then the projection of $\vec{v}$ onto L is given by

$$\text{Proj}_L(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \right) \vec{x}$$

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$$\text{Proj}_L(\vec{v}) = \frac{\begin{bmatrix}0 & -1 & 2\end{bmatrix} \cdot \begin{bmatrix}-2\\1\\3\end{bmatrix}}{\begin{bmatrix}-2 & 1 & 3\end{bmatrix} \cdot \begin{bmatrix}-2\\1\\3\end{bmatrix}} \cdot \begin{bmatrix}-2\\1\\3\end{bmatrix} = \frac{0(-2) - 1(1) + 2(3)}{-2(-2) + 1(1) + 3(3)} \cdot \begin{bmatrix}-2\\1\\3\end{bmatrix}$$

$$\text{Proj}_L(\vec{v}) = \frac{5}{14} \cdot \begin{bmatrix}-2\\1\\3\end{bmatrix} = \begin{bmatrix}-\frac{5}{7}\\\frac{5}{14}\\\frac{15}{14}\end{bmatrix}$$

■ 6. Find the projection of $\vec{v}$ onto L and the vector complement of $\vec{v}$ orthogonal to L.

$$L = \left\{ c \begin{bmatrix}-4\\0\\-1\end{bmatrix} \;\middle|\; c \in \mathbb{R} \right\}$$

$$\vec{v} = \begin{bmatrix}2\\-1\\1\end{bmatrix}$$

Solution:

The line L is given as all the scaled versions of the vector $\vec{x} = (-4,0,-1)$. Then the projection of $\vec{v}$ onto L is given by

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$$\text{Proj}_L(\vec{v}) = \left(\frac{\vec{v} \cdot \vec{x}}{\vec{x} \cdot \vec{x}}\right)\vec{x}$$

$$\text{Proj}_L(\vec{v}) = \frac{\begin{bmatrix}2 & -1 & 1\end{bmatrix} \cdot \begin{bmatrix}-4\\0\\-1\end{bmatrix}}{\begin{bmatrix}-4 & 0 & -1\end{bmatrix} \cdot \begin{bmatrix}-4\\0\\-1\end{bmatrix}} \cdot \begin{bmatrix}-4\\0\\-1\end{bmatrix} = \frac{2(-4) - 1(0) + 1(-1)}{-4(-4) + 0(0) - 1(-1)} \cdot \begin{bmatrix}-4\\0\\-1\end{bmatrix}$$

$$\text{Proj}_L(\vec{v}) = \frac{-9}{17} \cdot \begin{bmatrix}-4\\0\\-1\end{bmatrix} = \begin{bmatrix}\frac{36}{17}\\0\\\frac{9}{17}\end{bmatrix}$$

The vector complement of $\vec{v}$ orthogonal to L is the vector $\vec{v} - \text{Proj}_L(\vec{v})$.

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix}2\\-1\\1\end{bmatrix} - \begin{bmatrix}\frac{36}{17}\\0\\\frac{9}{17}\end{bmatrix}$$

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix}2 - \frac{36}{17}\\-1 - 0\\1 - \frac{9}{17}\end{bmatrix}$$

$$\vec{v} - \text{Proj}_L(\vec{v}) = \begin{bmatrix}-\frac{2}{17}\\-1\\\frac{8}{17}\end{bmatrix}$$

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COMPOSITIONS OF LINEAR TRANSFORMATIONS

1. If $S : \mathbb{R}^2 \to \mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$, then what is $T(S(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix} -x_2 + 3x_1 \\ x_1 + 2x_2 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} x_2 \\ x_1 - 3x_2 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

Solution:

Apply the transformation S to each column of the I_2 identity matrix.

$$S\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} -0 + 3(1) \\ 1 + 2(0) \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 + 3(0) \\ 0 + 2(1) \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

So the transformation S can be written as

$$S(\vec{x}) = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix} \vec{x}$$

Apply the transformation T to each column of the I_2 identity matrix.

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$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}0\\1-3(0)\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}1\\0-3(1)\end{bmatrix} = \begin{bmatrix}1\\-3\end{bmatrix}$$

So the transformation T can be written as

$$T(\vec{x}) = \begin{bmatrix}0 & 1\\1 & -3\end{bmatrix}\vec{x}$$

Then the composition $T \circ S$ can be written as

$$T(S(\vec{x})) = \begin{bmatrix}0 & 1\\1 & -3\end{bmatrix}\begin{bmatrix}3 & -1\\1 & 2\end{bmatrix}\vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix}0(3)+1(1) & 0(-1)+1(2)\\1(3)-3(1) & 1(-1)-3(2)\end{bmatrix}\vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix}0+1 & 0+2\\3-3 & -1-6\end{bmatrix}\vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix}1 & 2\\0 & -7\end{bmatrix}\vec{x}$$

Transform $\vec{x} = (-2,4)$.

$$T\left(S\left(\begin{bmatrix}-2\\4\end{bmatrix}\right)\right) = \begin{bmatrix}1 & 2\\0 & -7\end{bmatrix}\begin{bmatrix}-2\\4\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}-2\\4\end{bmatrix}\right)\right) = \begin{bmatrix}1(-2)+2(4)\\0(-2)-7(4)\end{bmatrix}$$

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$$T\left(S\left(\begin{bmatrix}-2\\4\end{bmatrix}\right)\right) = \begin{bmatrix}-2+8\\0-28\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}-2\\4\end{bmatrix}\right)\right) = \begin{bmatrix}6\\-28\end{bmatrix}$$

Therefore, we can say that the vector $\vec{x} = (-2,4)$ in $\mathbb{R}^2$ is transformed into the vector $\vec{z} = (6, -28)$ in $\mathbb{R}^2$.

■ 2. If $S : \mathbb{R}^2 \to \mathbb{R}^2$ and $T : \mathbb{R}^2 \to \mathbb{R}^2$, then what is $S(T(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix}-2x_1 + x_2\\-x_1 - x_2\end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix}-x_1 + 3x_2\\-2x_1 + 2x_2\end{bmatrix}$$

$$\vec{x} = \begin{bmatrix}-1\\2\end{bmatrix}$$

Solution:

Apply the transformation S to each column of the I_2 identity matrix.

$$S\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-2(1)+0\\-1-0\end{bmatrix} = \begin{bmatrix}-2\\-1\end{bmatrix}$$

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$$S\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-2(0)+1\\-0-1\end{bmatrix} = \begin{bmatrix}1\\-1\end{bmatrix}$$

So the transformation S can be written as

$$S(\vec{x}) = \begin{bmatrix}-2 & 1\\-1 & -1\end{bmatrix}\vec{x}$$

Apply the transformation T to each column of the I_2 identity matrix.

$$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-1+3(0)\\-2(1)+2(0)\end{bmatrix} = \begin{bmatrix}-1\\-2\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-0+3(1)\\-2(0)+2(1)\end{bmatrix} = \begin{bmatrix}3\\2\end{bmatrix}$$

So the transformation T can be written as

$$T(\vec{x}) = \begin{bmatrix}-1 & 3\\-2 & 2\end{bmatrix}\vec{x}$$

Then the composition $S \circ T$ can be written as

$$S(T(\vec{x})) = \begin{bmatrix}-2 & 1\\-1 & -1\end{bmatrix}\begin{bmatrix}-1 & 3\\-2 & 2\end{bmatrix}\vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix}-2(-1)+1(-2) & -2(3)+1(2)\\-1(-1)-1(-2) & -1(3)-1(2)\end{bmatrix}\vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix}2-2 & -6+2\\1+2 & -3-2\end{bmatrix}\vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix}0 & -4\\3 & -5\end{bmatrix}\vec{x}$$

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Transform $\vec{x} = (-1,2)$.

$$S\left(T\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)\right) = \begin{bmatrix}0 & -4\\3 & -5\end{bmatrix}\begin{bmatrix}-1\\2\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)\right) = \begin{bmatrix}0(-1)-4(2)\\3(-1)-5(2)\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)\right) = \begin{bmatrix}0-8\\-3-10\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}-1\\2\end{bmatrix}\right)\right) = \begin{bmatrix}-8\\-13\end{bmatrix}$$

Therefore, we can say that the vector $\vec{x} = (-1,2)$ in $\mathbb{R}^2$ is transformed into the vector $\vec{z} = (-8,-13)$ in $\mathbb{R}^2$.

■ 3. If $S : \mathbb{R}^3 \to \mathbb{R}^3$ and $T : \mathbb{R}^3 \to \mathbb{R}^3$, then what is $T(S(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix}-2x_1 + x_2 - x_3\\-x_2 + x_3\\x_1 + 2x_2\end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix}-3x_3\\x_1 + x_2 - 2x_3\\-x_1 - x_2 + x_3\end{bmatrix}$$

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$$\vec{x} = \begin{bmatrix} -2 \\ 1 \\ -3 \end{bmatrix}$$

Solution:

Apply the transformation S to each column of the I_3 identity matrix.

$$S\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -2(1) + 0 - 0 \\ -0 + 0 \\ 1 + 2(0) \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -2(0) + 1 - 0 \\ -1 + 0 \\ 0 + 2(1) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -2(0) + 0 - 1 \\ -0 + 1 \\ 0 + 2(0) \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

So the transformation S can be written as

$$S(\vec{x}) = \begin{bmatrix} -2 & 1 & -1 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \vec{x}$$

Apply the transformation T to each column of the I_3 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -3(0) \\ 1 + 0 - 2(0) \\ -1 - 0 + 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

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$$T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right)=\begin{bmatrix}-3(0)\\0+1-2(0)\\-0-1+0\end{bmatrix}=\begin{bmatrix}0\\1\\-1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right)=\begin{bmatrix}-3(1)\\0+0-2(1)\\-0-0+1\end{bmatrix}=\begin{bmatrix}-3\\-2\\1\end{bmatrix}$$

So the transformation T can be written as

$$T(\vec{x})=\begin{bmatrix}0&0&-3\\1&1&-2\\-1&-1&1\end{bmatrix}\vec{x}$$

Then the composition $T\circ S$ can be written as

$$T(S(\vec{x}))=\begin{bmatrix}0&0&-3\\1&1&-2\\-1&-1&1\end{bmatrix}\begin{bmatrix}-2&1&-1\\0&-1&1\\1&2&0\end{bmatrix}\vec{x}$$

$$T(S(\vec{x}))=\begin{bmatrix}0(-2)+0(0)-3(1)&0(1)+0(-1)-3(2)&0(-1)+0(1)-3(0)\\1(-2)+1(0)-2(1)&1(1)+1(-1)-2(2)&1(-1)+1(1)-2(0)\\-1(-2)-1(0)+1(1)&-1(1)-1(-1)+1(2)&-1(-1)-1(1)+1(0)\end{bmatrix}\vec{x}$$

$$T(S(\vec{x}))=\begin{bmatrix}0+0-3&0+0-6&0+0+0\\-2+0-2&1-1-4&-1+1-0\\2-0+1&-1+1+2&1-1+0\end{bmatrix}\vec{x}$$

$$T(S(\vec{x}))=\begin{bmatrix}-3&-6&0\\-4&-4&0\\3&2&0\end{bmatrix}\vec{x}$$

Transform $\vec{x}=(-2,1,-3)$.

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$$T\left(S\left(\begin{bmatrix}-2\\1\\-3\end{bmatrix}\right)\right) = \begin{bmatrix}-3 & -6 & 0\\-4 & -4 & 0\\3 & 2 & 0\end{bmatrix}\begin{bmatrix}-2\\1\\-3\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}-2\\1\\-3\end{bmatrix}\right)\right) = \begin{bmatrix}-3(-2)-6(1)+0(-3)\\-4(-2)-4(1)+0(-3)\\3(-2)+2(1)+0(-3)\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}-2\\1\\-3\end{bmatrix}\right)\right) = \begin{bmatrix}6-6+0\\8-4+0\\-6+2+0\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}-2\\1\\-3\end{bmatrix}\right)\right) = \begin{bmatrix}0\\4\\-4\end{bmatrix}$$

Therefore, we can say that the vector $\vec{x} = (-2,1,-3)$ in $\mathbb{R}^3$ is transformed into the vector $\vec{z} = (0,4,-4)$ in $\mathbb{R}^3$.

4. If $S : \mathbb{R}^2 \to \mathbb{R}^3$ and $T : \mathbb{R}^3 \to \mathbb{R}^2$, then what is $S(T(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix}-x_1\\x_1-3x_2\\2x_2-3x_1\end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix}2x_1-x_3\\x_2-x_1+x_3\end{bmatrix}$$

$$\vec{x} = \begin{bmatrix}1\\-2\\-1\end{bmatrix}$$

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Solution:

Apply the transformation S to each column of the I_2 identity matrix.

$$S\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-1\\1-3(0)\\2(0)-3(1)\end{bmatrix} = \begin{bmatrix}-1\\1\\-3\end{bmatrix}$$

$$S\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-0\\0-3(1)\\2(1)-3(0)\end{bmatrix} = \begin{bmatrix}0\\-3\\2\end{bmatrix}$$

So the transformation S can be written as

$$S(\vec{x}) = \begin{bmatrix}-1 & 0\\1 & -3\\-3 & 2\end{bmatrix}\vec{x}$$

Apply the transformation T to each column of the I_3 identity matrix.

$$T\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}2(1)-0\\0-1+0\end{bmatrix} = \begin{bmatrix}2\\-1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = \begin{bmatrix}2(0)-0\\1-0+0\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = \begin{bmatrix}2(0)-1\\0-0+1\end{bmatrix} = \begin{bmatrix}-1\\1\end{bmatrix}$$

So the transformation T can be written as

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$$T(\vec{x}) = \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix} \vec{x}$$

Then the composition $S \circ T$ can be written as

$$S(T(\vec{x})) = \begin{bmatrix} -1 & 0 \\ 1 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} -1(2)+0(-1) & -1(0)+0(1) & -1(-1)+0(1) \\ 1(2)-3(-1) & 1(0)-3(1) & 1(-1)-3(1) \\ -3(2)+2(-1) & -3(0)+2(1) & -3(-1)+2(1) \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} -2+0 & 0+0 & 1+0 \\ 2+3 & 0-3 & -1-3 \\ -6-2 & 0+2 & 3+2 \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} -2 & 0 & 1 \\ 5 & -3 & -4 \\ -8 & 2 & 5 \end{bmatrix} \vec{x}$$

Transform $\vec{x} = (1, -2, -1)$.

$$S\left(T\left(\begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} -2 & 0 & 1 \\ 5 & -3 & -4 \\ -8 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} -2(1)+0(-2)+1(-1) \\ 5(1)-3(-2)-4(-1) \\ -8(1)+2(-2)+5(-1) \end{bmatrix}$$

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$$S\left(T\left(\begin{bmatrix}1\\-2\\-1\end{bmatrix}\right)\right) = \begin{bmatrix}-2+0-1\\5+6+4\\-8-4-5\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}1\\-2\\-1\end{bmatrix}\right)\right) = \begin{bmatrix}-3\\15\\-17\end{bmatrix}$$

Therefore, we can say that the vector $\vec{x} = (1, -2, -1)$ in $\mathbb{R}^3$ is transformed into the vector $\vec{z} = (-3, 15, -17)$ in $\mathbb{R}^3$.

■ 5. If $S : \mathbb{R}^2 \to \mathbb{R}^3$ and $T : \mathbb{R}^3 \to \mathbb{R}^2$, then what is $T(S(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix}-x_1 + x_2\\2x_2 - 3x_1\\x_1 + 2x_2\end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix}x_1 - 2x_2 + x_3\\x_1 + x_2 - x_3\end{bmatrix}$$

$$\vec{x} = \begin{bmatrix}-2\\-1\end{bmatrix}$$

Solution:

Apply the transformation S to each column of the I_2 identity matrix.

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$$S\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = \begin{bmatrix}-1+0\\2(0)-3(1)\\1+2(0)\end{bmatrix} = \begin{bmatrix}-1\\-3\\1\end{bmatrix}$$

$$S\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = \begin{bmatrix}-0+1\\2(1)-3(0)\\0+2(1)\end{bmatrix} = \begin{bmatrix}1\\2\\2\end{bmatrix}$$

So the transformation S can be written as

$$S(\vec{x}) = \begin{bmatrix}-1 & 1\\-3 & 2\\1 & 2\end{bmatrix}\vec{x}$$

Apply the transformation T to each column of the I_3 identity matrix.

$$T\left(\begin{bmatrix}1\\0\\0\end{bmatrix}\right) = \begin{bmatrix}1-2(0)+0\\1+0-0\end{bmatrix} = \begin{bmatrix}1\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\1\\0\end{bmatrix}\right) = \begin{bmatrix}0-2(1)+0\\0+1-0\end{bmatrix} = \begin{bmatrix}-2\\1\end{bmatrix}$$

$$T\left(\begin{bmatrix}0\\0\\1\end{bmatrix}\right) = \begin{bmatrix}0-2(0)+1\\0+0-1\end{bmatrix} = \begin{bmatrix}1\\-1\end{bmatrix}$$

So the transformation T can be written as

$$T(\vec{x}) = \begin{bmatrix}1 & -2 & 1\\1 & 1 & -1\end{bmatrix}\vec{x}$$

Then the composition $T \circ S$ can be written as

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$$T(S(\vec{x})) = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -3 & 2 \\ 1 & 2 \end{bmatrix} \vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix} 1(-1) - 2(-3) + 1(1) & 1(1) - 2(2) + 1(2) \\ 1(-1) + 1(-3) - 1(1) & 1(1) + 1(2) - 1(2) \end{bmatrix} \vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix} -1 + 6 + 1 & 1 - 4 + 2 \\ -1 - 3 - 1 & 1 + 2 - 2 \end{bmatrix} \vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} \vec{x}$$

Transform $\vec{x} = (-2, -1)$.

$$T\left(S\left(\begin{bmatrix} -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} 6 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix} -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} 6(-2) - 1(-1) \\ -5(-2) + 1(-1) \end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix} -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} -12 + 1 \\ 10 - 1 \end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix} -2 \\ -1 \end{bmatrix}\right)\right) = \begin{bmatrix} -11 \\ 9 \end{bmatrix}$$

Therefore, we can say that the vector $\vec{x} = (-2, -1)$ in $\mathbb{R}^2$ is transformed into the vector $\vec{z} = (-11,9)$ in $\mathbb{R}^2$.

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6. If $S : \mathbb{R}^3 \to \mathbb{R}^3$ and $T : \mathbb{R}^3 \to \mathbb{R}^3$, then what are $T(S(\vec{x}))$ and $S(T(\vec{x}))$?

$$S(\vec{x}) = \begin{bmatrix} -x_3 + 2x_2 \\ x_1 - x_3 \\ x_1 \end{bmatrix}$$

$$T(\vec{x}) = \begin{bmatrix} -2x_1 + x_2 + 2x_3 \\ 3x_1 \\ x_1 - 2x_2 + x_3 \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$$

Solution:

Apply the transformation S to each column of the I_3 identity matrix.

$$S\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -0 + 2(0) \\ 1 - 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -0 + 2(1) \\ 0 - 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$S\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 + 2(0) \\ 0 - 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

So the transformation S can be written as

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$$S(\vec{x}) = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \vec{x}$$

Apply the transformation T to each column of the I_3 identity matrix.

$$T\left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -2(1) + 0 + 2(0) \\ 3(1) \\ 1 - 2(0) + 0 \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} -2(0) + 1 + 2(0) \\ 3(0) \\ 0 - 2(1) + 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$T\left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -2(0) + 0 + 2(1) \\ 3(0) \\ 0 - 2(0) + 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

So the transformation T can be written as

$$T(\vec{x}) = \begin{bmatrix} -2 & 1 & 2 \\ 3 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \vec{x}$$

Then the composition $T \circ S$ can be written as

$$T(S(\vec{x})) = \begin{bmatrix} -2 & 1 & 2 \\ 3 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix} -2(0) + 1(1) + 2(1) & -2(2) + 1(0) + 2(0) & -2(-1) + 1(-1) + 2(0) \\ 3(0) + 0(1) + 0(1) & 3(2) + 0(0) + 0(0) & 3(-1) + 0(-1) + 0(0) \\ 1(0) - 2(1) + 1(1) & 1(2) - 2(0) + 1(0) & 1(-1) - 2(-1) + 1(0) \end{bmatrix} \vec{x}$$

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$$T(S(\vec{x})) = \begin{bmatrix} 0+1+2 & -4+0+0 & 2-1+0 \\ 0+0+0 & 6+0+0 & -3+0+0 \\ 0-2+1 & 2-0+0 & -1+2+0 \end{bmatrix} \vec{x}$$

$$T(S(\vec{x})) = \begin{bmatrix} 3 & -4 & 1 \\ 0 & 6 & -3 \\ -1 & 2 & 1 \end{bmatrix} \vec{x}$$

And the composition $S \circ T$ can be written as

$$S(T(\vec{x})) = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -2 & 1 & 2 \\ 3 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} 0(-2)+2(3)-1(1) & 0(1)+2(0)-1(-2) & 0(2)+2(0)-1(1) \\ 1(-2)+0(3)-1(1) & 1(1)+0(0)-1(-2) & 1(2)+0(0)-1(1) \\ 1(-2)+0(3)+0(1) & 1(1)+0(0)+0(-2) & 1(2)+0(0)+0(1) \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} 0+6-1 & 0+0+2 & 0+0-1 \\ -2+0-1 & 1+0+2 & 2+0-1 \\ -2+0+0 & 1+0+0 & 2+0+0 \end{bmatrix} \vec{x}$$

$$S(T(\vec{x})) = \begin{bmatrix} 5 & 2 & -1 \\ -3 & 3 & 1 \\ -2 & 1 & 2 \end{bmatrix} \vec{x}$$

Then $T(S(\vec{x}))$, where $\vec{x} = (0, -1, 3)$, is

$$T\left(S\left(\begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}\right)\right) = \begin{bmatrix} 3 & -4 & 1 \\ 0 & 6 & -3 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}\right)\right) = \begin{bmatrix} 3(0)-4(-1)+1(3) \\ 0(0)+6(-1)-3(3) \\ -1(0)+2(-1)+1(3) \end{bmatrix}$$

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$$T\left(S\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}0+4+3\\0-6-9\\0-2+3\end{bmatrix}$$

$$T\left(S\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}7\\-15\\1\end{bmatrix}$$

And $S(T(\vec{x}))$, where $\vec{x} = (0, -1, 3)$, is

$$S\left(T\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}5 & 2 & -1\\-3 & 3 & 1\\-2 & 1 & 2\end{bmatrix}\begin{bmatrix}0\\-1\\3\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}5(0)+2(-1)-1(3)\\-3(0)+3(-1)+1(3)\\-2(0)+1(-1)+2(3)\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}0-2-3\\0-3+3\\0-1+6\end{bmatrix}$$

$$S\left(T\left(\begin{bmatrix}0\\-1\\3\end{bmatrix}\right)\right) = \begin{bmatrix}-5\\0\\5\end{bmatrix}$$

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